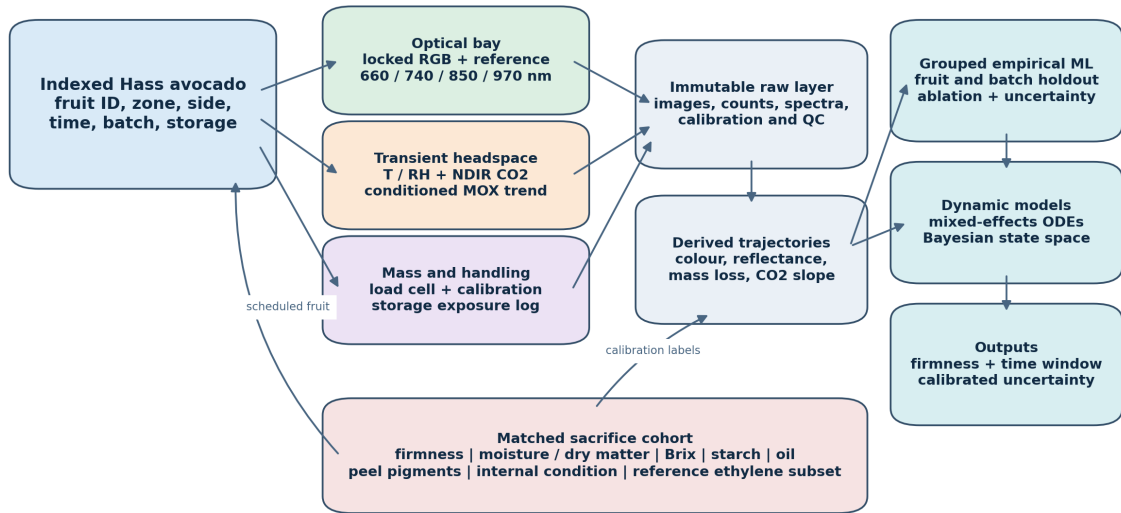


Development of a Multimodal Non-Destructive Avocado Ripeness Characterization System

Optical spectroscopy, embedded sensing, machine learning, and scientific machine learning

Proposed multimodal avocado characterization system

Repeated non-destructive measurements linked to matched destructive ground truth



Claims boundary: RGB, NIR, mass, CO2, and MOX are observations. Internal chemistry is a calibrated prediction only when matched reference assays support it.

478 independent fruits	14,710 audited RGB images	44 BOM line items	US\$723.62 MVP planning total
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Prepared 30 July 2026

For discussion with collaborators, supervisors, and prospective laboratory or funding partners

Evidence boundary. Results reproduced here are internal fruit-grouped analyses of one public study. Hardware, equations, experimental endpoints, and scientific-ML methods are proposed until prospectively validated.

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Development of a Multimodal Non-Destructive Avocado Ripeness Characterization System

Optical spectroscopy, embedded sensing, machine learning, and scientific machine learning

Proposal status: integrated research and prototype design **Prepared:** 30 July 2026 **Primary cultivar:** Hass avocado (*Persea americana* Mill.) **Intended discussion:** Collaborators, academic supervisors, and potential laboratory or funding partners

Evidence and claim convention

This proposal deliberately separates five kinds of statement:

- **DATA-DEMONSTRATED:** directly reproduced from files inspected in this workspace.
- **LITERATURE-SUPPORTED:** reported by a cited primary source but not independently reproduced here.
- **INFERRED:** a reasoned interpretation of available evidence.
- **PROPOSED:** a method, apparatus, model, or decision rule for future validation.
- **NOT IDENTIFIABLE:** current data cannot determine the requested quantity or causal effect.

The distinction matters because a large image collection can still contain only a small number of independent fruits, a sensor feature can be predictive without being chemically specific, and a mathematical equation can be plausible without its parameters being identifiable.

1. Executive summary

Avocado quality changes rapidly after harvest and is difficult to assess without cutting the fruit. Consumers and retailers normally combine peel colour, handling history, and manual squeezing, while research laboratories can measure firmness, dry matter, moisture, oil, respiration gases, pigments, and spectra. The inexpensive observations are subjective and incomplete; the more specific measurements are destructive or require equipment that is not practical at point of use. An inaccurate readiness estimate contributes to rejected fruit, premature consumption, missed eating windows, and avoidable waste.

Published work establishes that several non-destructive channels contain useful information, but it does not yet provide a universally validated avocado "ripeness meter." RipeTrack reconstructed spectral information from mobile RGB/NIR observations and reported strong ripeness and remaining-life results, but its avocado subset contained only 13 independent fruits and used ethylene-trajectory-derived labels ([Waseem, Sharma, and Hefeeda](#)). Davur et al. reported low hyperspectral age-prediction error, but their sub-images from each source image were allocated across training, validation, and test partitions, so the published result does not demonstrate generalization to unseen fruit ([Davur et al.](#)). Clark et al. showed that NIR interactance can predict avocado dry matter, while later work has explored moisture, oil, colour, firmness, and deep image models ([Clark et al.00046-2](#)); [Olawejaju et al.](#); [Lee, Li, and Ma](#)). These studies support feasibility, but their endpoints, spectra, cohorts, split units, devices, and access conditions differ.

This project proposes a controlled, multimodal, fruit-grouped study that joins:

1. locked and calibrated RGB images;
2. repeatable reflectance or interactance measurements, first at 660, 740, 850,

and 970 nm and then, where resources permit, over 900-1700 or 900-2500 nm;

3. fruit mass, chamber temperature, relative humidity, and airflow;
4. standardized transient headspace CO₂ and broad VOC observations, with reference ethylene measurement in a research-grade subset;
5. destructive firmness, moisture/dry matter, soluble solids, starch, oil, peel pigments, and internal-condition reference measurements in matched sacrifice cohorts.

The computational work completed for this proposal demonstrates the importance of the validation design. The complete, CC BY 4.0 Hass Avocado Ripening Photographic Dataset was downloaded and audited ([Xavier, Rodrigues, and Silva](#)). It contains 14,710 available photographs but only 478 biological fruits. Five-fold fruit-grouped validation of compact colour features produced 0.731 balanced accuracy for the best five-stage classifier and a 1.853-day mean absolute error for a retrospective days-to-stage-4 target. When an entire storage condition was held out, balanced accuracy fell as low as 0.497. These are **DATA-DEMONSTRATED** internal results, not deployment claims. They show both that peel colour is useful and that its relationship with stage shifts across conditions.

The proposal therefore treats ripening as a partially observed dynamic system rather than as a single image label. Conventional baselines will be compared with mixed-effects, state-space, survival, chemometric, and temporal machine learning models. Candidate mass-transfer, pigment, softening, respiration, and ethylene equations will be fitted only to variables that are measured. Bayesian state-space modeling and nonlinear mixed-effects ODEs are the preferred first scientific-machine-learning methods; SINDy, symbolic regression, Neural ODEs, and physics-informed neural networks are conditional extensions, not automatic claims of discovered biology.

The minimum viable apparatus has an audited planning total of US\$723.62, including allowances. A research configuration that assumes access to shared core instruments has a planned purchased-component total of US\$6,591.51. A US\$43,466.01 buy-new figure is retained only as a non-quotation planning proxy for major shared instruments. Exact components, manufacturer part numbers, sources, limitations, and price status are recorded in Appendix D and the machine-readable bill of materials.

The intended output is not merely a high classification score. It is an evidence-auditable dataset and calibrated prototype that can answer which sensors contribute independent information, how environmental conditions change transition rates, how uncertainty grows under domain shift, and which biological quantities remain impossible to infer from a low-cost device.

2. Research questions, objectives, and hypotheses

2.1 Primary research question

PROPOSED: Can calibrated longitudinal RGB, NIR, mass, environmental, and respiration observations predict instrumental avocado firmness and time to a predeclared ready-to-eat or end-of-acceptable-quality endpoint for unseen fruits and unseen harvest batches more accurately and more reliably than RGB-only or single-sensor baselines?

The question has four deliberate restrictions:

- "calibrated" requires dark/white/colour/gas/mass controls;
- "longitudinal" requires repeated observations of the same identified fruit;
- "unseen" requires fruit- and batch-level holdouts;
- "predeclared endpoint" prevents a threshold from being chosen after seeing the test results.

2.2 Secondary research questions

1. Which channels add incremental value after controlled RGB colour is known?
2. Does 900-1700 nm or 900-2500 nm spectroscopy materially improve prediction of moisture/dry matter and firmness over the four-band MVP?

3. How do temperature, RH, airflow, pre-storage history, harvest batch, and initial dry matter modify rates of mass loss, softening, pigment change, and CO₂ accumulation?
4. Can a coupled state-space or kinetic model improve temporal consistency, uncertainty calibration, and transfer across storage regimes relative to a black-box predictor?
5. Can low-cost CO₂ and MOX channels contribute predictive information without being misrepresented as compound-specific ethylene or ethanol measurements?
6. At what measurement frequency does the information gain cease to justify the extra handling and apparatus time?
7. How early can the system issue a useful forecast, and how does forecast uncertainty change as the endpoint approaches?

2.3 Objectives

- Build and metrologically qualify a repeatable scanning/headspace apparatus.
- Publish a fruit-resolved longitudinal data contract with immutable raw files, calibration records, and destructive reference links.
- Establish leakage-resistant optical and multimodal baselines.
- Quantify sensor contribution through prespecified ablation.
- Estimate environmental and biological rate parameters with hierarchical uncertainty.
- Compare empirical and mechanistic models on the same external batches.
- State clear non-claims where specificity or identifiability is absent.

2.4 Hypotheses and null hypotheses

ID	Alternative hypothesis	Null hypothesis	Primary test
H1	Multimodal sensing lowers batch-held-out firmness MAE relative to calibrated RGB alone.	The paired error difference is zero or favors RGB alone.	Batch-level paired bootstrap or hierarchical error contrast.
H2	Longitudinal features lower remaining-time error relative to a single current observation.	Trajectory history provides no out-of-batch benefit.	Nested grouped comparison with identical outer folds.
H3	Research-grade NIR predicts moisture/dry matter after fruit- and batch-grouped calibration.	NIR does not outperform a training-set mean or initial-mass baseline.	External RMSEP, bias, R-squared, and RPD with uncertainty.
H4	Calibrated colour adds information about peel pigment/state but is insufficiently invariant across storage regimes.	Colour is either non-predictive or fully invariant.	Ablation plus storage/batch/device holdouts and interaction tests.
H5	Temperature and RH modify ripening and water-loss rate parameters.	Rate parameters do not differ after batch/chamber adjustment.	Hierarchical kinetic model and likelihood/posterior contrasts.
H6	A dynamic state-space model improves forecast calibration or temporal consistency without unacceptable accuracy loss.	It offers no improvement over a matched black-box baseline.	Predeclared composite of error, interval coverage, and monotonicity violations.
H7	CO ₂ accumulation and conditioned MOX trajectories add information beyond optical and mass channels.	Their conditional permutation/ablation contribution is zero.	Outer-fold ablation with fruit-cluster uncertainty.
H8	A low-cost indicated-ethylene channel is not quantitatively interchangeable with a reference analyzer in mixed avocado headspace.	Agreement is within predeclared analytical limits across stage, T, RH, CO ₂ , and ethanol challenge.	Method-comparison study, bias, LoA, drift, response/recovery.

No hypothesis treats statistical significance as practical usefulness. Minimum practically/commercially meaningful differences must be fixed with stakeholders before confirmatory analysis.

3. State of the evidence and research gap

3.1 Smartphones, RGB, and colour

Hass peel normally loses green colour and becomes purple-black during ripening, making controlled colour a useful low-cost channel. The process is not simply "chlorophyll becomes black pigment." Chlorophyll a and b decline mainly early, while cyanidin-3-O-glucoside accumulation contributes later darkening ([Cox et al.](#)). Temperature can alter anthocyanin accumulation and visible darkening without an equivalent effect on chlorophyll loss ([Sibeko et al.](#)). Colour and internal softening can therefore desynchronize.

Smartphone studies demonstrate useful predictive associations. Cho et al. reported an artificial-neural-network relationship between smartphone colour features and avocado maturity, and Lee, Li, and Ma used 1,400 iPhone images from 140 avocados to predict firmness and classify internal freshness ([Cho et al.](#); [Lee, Li, and Ma](#)). However, random image splits can place views of the same fruit in training and test data. Cross-batch results in the latter paper are more relevant and show directional domain shift. Colour is retained here as a required baseline and ablation channel, not accepted as universal internal ground truth.

3.2 NIR and hyperspectral sensing

NIR measurements mix absorption and scattering. Water has strong bands near 1450 and 1930 nm and a weaker feature near 970 nm; C-H combinations and overtones associated with lipid and other organic material appear in several regions including approximately 900-920, 1200, 1700, and 2300 nm. A four-band silicon-detector prototype that stops near 1100 nm cannot separate water, lipid, scattering, and skin effects with laboratory specificity. The assignments and their overlap are supported by avocado NIR work spanning silicon and SWIR ranges ([Clark et al.00046-2](#); [Olarewaju et al.](#); [Mishra et al.](#)).

Clark et al. reported strong dry-matter prediction using NIR interactance in Hass avocados, with an independent prediction result of approximately R-squared 0.88 and RMSEP 1.8 percentage points in that study ([Clark et al.00046-2](#)). Olarewaju et al. evaluated broader NIR for moisture, dry matter, and oil and found stronger moisture/dry-matter than oil performance ([Olarewaju et al.](#)). Melado-Herreros et al. combined flesh firmness and dry matter in a proposed ripening index and used 380-2000 nm Vis-NIR spectroscopy; their analysis identified potential for a lower-cost 400-1100 nm implementation ([Melado-Herreros et al.](#)). That finding supports investigating a silicon-range device, but a four-discrete-band MVP contains much less spectral information and must be benchmarked rather than assumed equivalent.

Mishra et al. directly tested avocado firmness across dehydration conditions using 350-2500 nm Vis-NIR, acoustic sensing, limited compression, and penetrometer references. Peel dehydration impaired Vis-NIR transfer, while Vis-NIR/acoustic fusion reduced firmness-prediction error by 21% in their study ([Mishra et al.](#)). This is especially important for the proposed design: water history is both a biological driver and a source of optical domain shift, and independent modalities can reduce - but not automatically eliminate - that confounding.

Davur et al. acquired 551 hyperspectral observations from 80 Hass fruits between 388.9 and 1005.3 nm, but their sub-image partition creates a leakage risk for unseen-fruit claims ([Davur et al.](#)).

RipeTrack is important because it joins hyperspectral proof-of-concept, ethylene-derived staging, RGB-to-spectrum reconstruction, and phone NIR. Its reported aggregate performance is promising, but its avocado experiments must be interpreted with the independent-fruit counts and label construction, not with image count alone ([Waseem, Sharma, and Hefeeda](#)).

3.3 Gases, respiration, and storage

Avocado is climacteric: respiration and ethylene participate in the ripening transition. A sensor that measures chamber CO₂ is not measuring ethylene, and a metal-oxide gas sensor is not measuring a unique VOC. Cross-sensitive electrochemical ethylene devices require matrix-specific comparison with a reference analyzer or gas chromatography.

The local Scientific Reports paper concerns apples rather than avocados. Bratu et al. followed 20 apples from four cultivars over 35 days and observed declining ethylene and increasing ethanol under their protocol. Its multispectral temporal patterns were mostly nonuniform or inconclusive; the paper should not be cited as showing that imaging strongly

tracked internal gas change ([Bratu et al.](#)). Its contribution here is methodological complementarity, not avocado-specific calibration.

Gwanpua et al. already developed an autocatalytic ethylene model coupled through Michaelis-Menten-type relationships to avocado softening and colour, with temperature dependence ([Gwanpua et al.](#)). Sierra et al. also modeled postharvest avocado quality kinetics ([Sierra et al.](#)). More recently, van der Sman, Guo, and Pedreschi applied a physics-informed neural network to postharvest avocado firmness using literature datasets and a physiological enzymatic-softening/metabolic-rate framework ([van der Sman et al.](#)). Their results also identify limits that matter here: differences between controlled-atmosphere and regular-air storage need further model structure, and orchard, climate, and harvest variation cannot be represented by a single initial state. Guo et al. subsequently published a broader scientific-machine-learning framework and model-selection guidance for fruit and vegetable quality prediction ([Guo et al.](#)). Hernández et al. provide another relevant dynamic precedent by using harvest-time metabolites as physiological markers for batch-specific Hass softening behavior across postharvest logistics ([Hernández et al.](#)).

The proposed project is therefore not the first governing-equation or avocado PINN model. Its prospective contribution would be a calibrated multimodal observation system, rigorous fruit/batch/device validation, matched chemistry, and a like-for-like comparison of statistical, mechanistic, and flexible dynamic models on newly collected data.

3.4 Chemistry and destructive reference measurements

The biologically relevant state is multi-component:

- cell-wall and pectin changes affect mechanical firmness;
- water exchange affects mass, fresh-basis concentration, and texture;
- respiration removes carbon and releases CO₂ and water;
- ethylene regulates climacteric responses;
- peel chlorophyll and anthocyanin trajectories affect visible colour;
- starch and soluble carbohydrates may change, while Brix remains a

concentration rather than a direct starch-conversion balance;

- lipid content is strongly tied to preharvest maturity, and an increase in

fresh-basis oil percentage after harvest can reflect water loss rather than new oil synthesis;

- internal bruising, vascular browning, rot, and chilling injury can break the

relationship between external appearance and acceptability.

Moisture percentage by oven or validated moisture analysis is destructive:

$$\mathrm{Moisture}_{\mathrm{wb}}(\%) = 100 \frac{m_{\mathrm{fresh}} - m_{\mathrm{dry}}}{m_{\mathrm{fresh}}}.$$

Dry matter on a wet basis is $100 - \mathrm{Moisture}_{\mathrm{wb}}$ when the same sample and method are used. Brix should be modeled jointly with water and, where starch conversion is claimed, a validated carbohydrate assay. A mass-balance interpretation is:

$$\Delta C_{\mathrm{soluble}} = \text{biochemical production/consumption} + \text{concentration or dilution from water change},$$

not "all Brix increase equals starch converted to sugar."

3.5 Gap this project addresses

No inspected public dataset jointly contains longitudinal calibrated RGB/NIR, mass, continuous environment, firmness, respiration gases, moisture, dry matter, oil, carbohydrate, pigments, and internal condition. Consequently the downloaded RGB data cannot identify internal water, oil, pigment concentration, gas kinetics, starch conversion, or a causal temperature coefficient. The proposed matched experiment is designed to make a subset of these relationships estimable while preserving honest limits on compound specificity.

Detailed paper summaries and corrected interpretations appear in Appendix A; the verified bibliography is in Appendix B.

4. Public datasets and completed computational audit

4.1 Acquisition decisions

Dataset/source	Access and license finding	Local action	Modeling decision
Hass Avocado Ripening Photographic Dataset v1	Public, CC BY 4.0	Complete 418,384,370-byte ZIP downloaded, hashed, extracted, and audited	Primary RGB/longitudinal benchmark
Kamat avocado/strawberry archive v1	Public, CC BY 4.0	Complete 2,110,769,149-byte ZIP downloaded and hashed; retained unextracted	Excluded until archive structure, originals, augmentation lineage, and fruit IDs are resolved
DeepHS Fruit	Public file index; no verified dataset license	Official repository and small annotations retained; 77.3-GB avocado payload not downloaded	License block; no scientific model fitted
RipeTrack	Public repository/Drive links; no verified dataset license	Official code repository retained; large fruit payloads not downloaded	License block; no scientific model fitted
Davur et al. HSI	Data unavailable due to IP arrangements	Paper audited	Literature only
Han et al.; Lee et al.	Reasonable-request access	Not represented as inspected	Literature only unless access and rights are granted

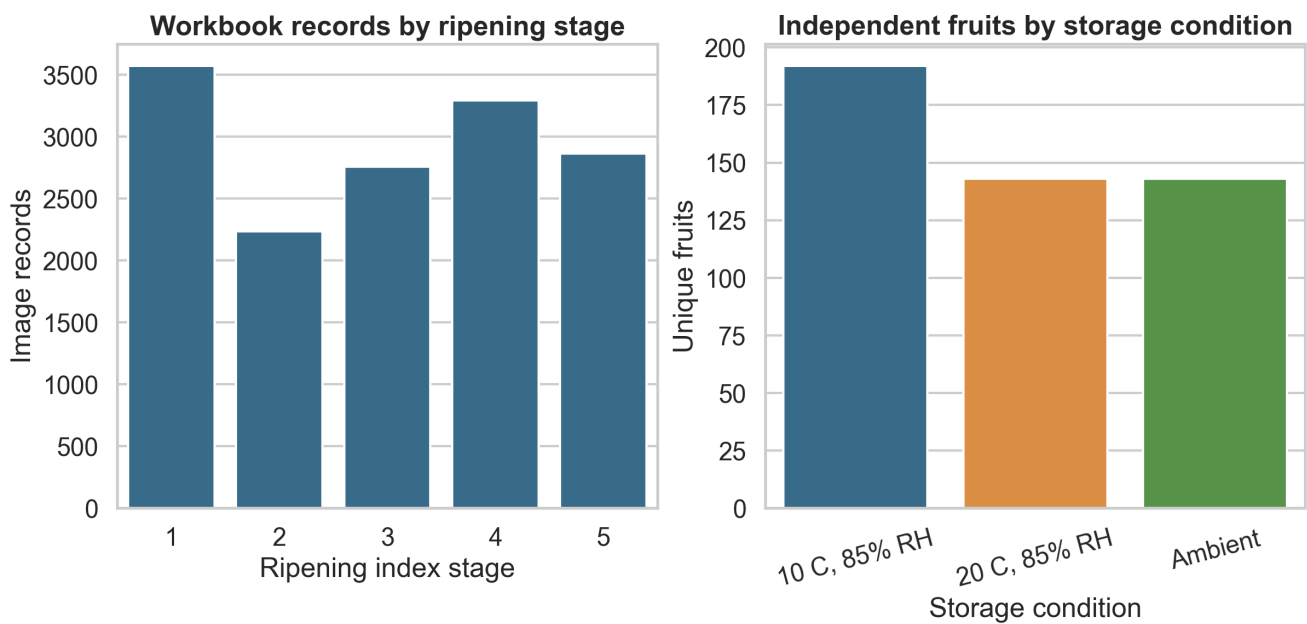
Public accessibility is not equivalent to permission to reuse or redistribute. The exact inventory, checksums, source URLs, status, and recommended use are in [Appendix C](#), [dataset_inventory.csv](#), and [dataset-audit.md](#).

4.2 Primary dataset quality

DATA-DEMONSTRATED:

Quantity	Result
Biological fruits	478
Workbook rows	14,722
Available JPEG images	14,710
Workbook rows missing a JPEG	12
Unlisted JPEGs	0
Duplicate workbook filenames	0
Filename/workbook metadata disagreements	0
Stage records 1/2/3/4/5	3,572 / 2,234 / 2,758 / 3,294 / 2,864
Fruits reaching stage 4 during follow-up	426
Fruits reaching stage 5 during follow-up	410

The median first recorded stage-4 day was 17 days in the 10 C/85% RH group, 7 days in the 20 C/85% RH group, and 7 days in the ambient group. These are descriptive associations, not a randomized causal temperature estimate.



Primary RGB dataset distribution by storage group and stage

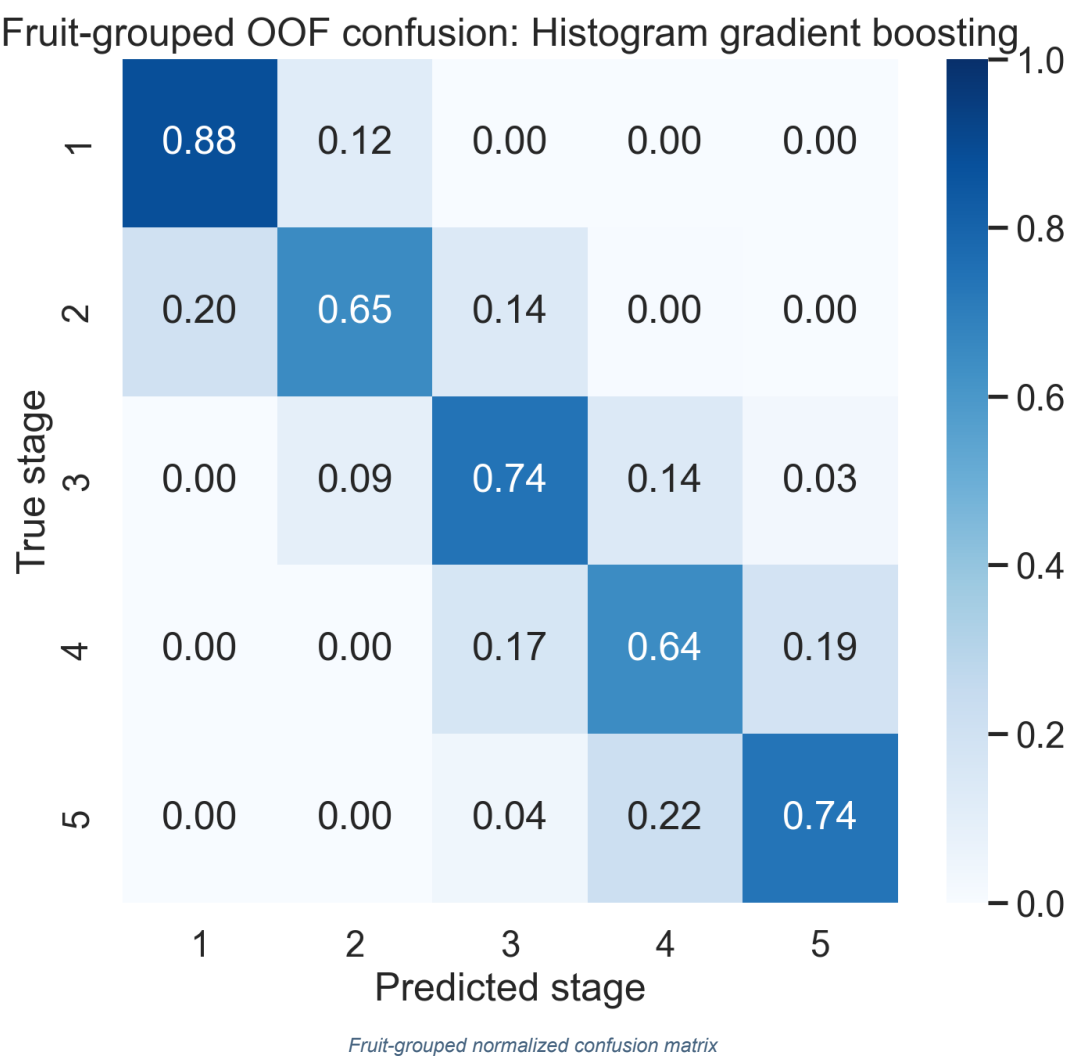
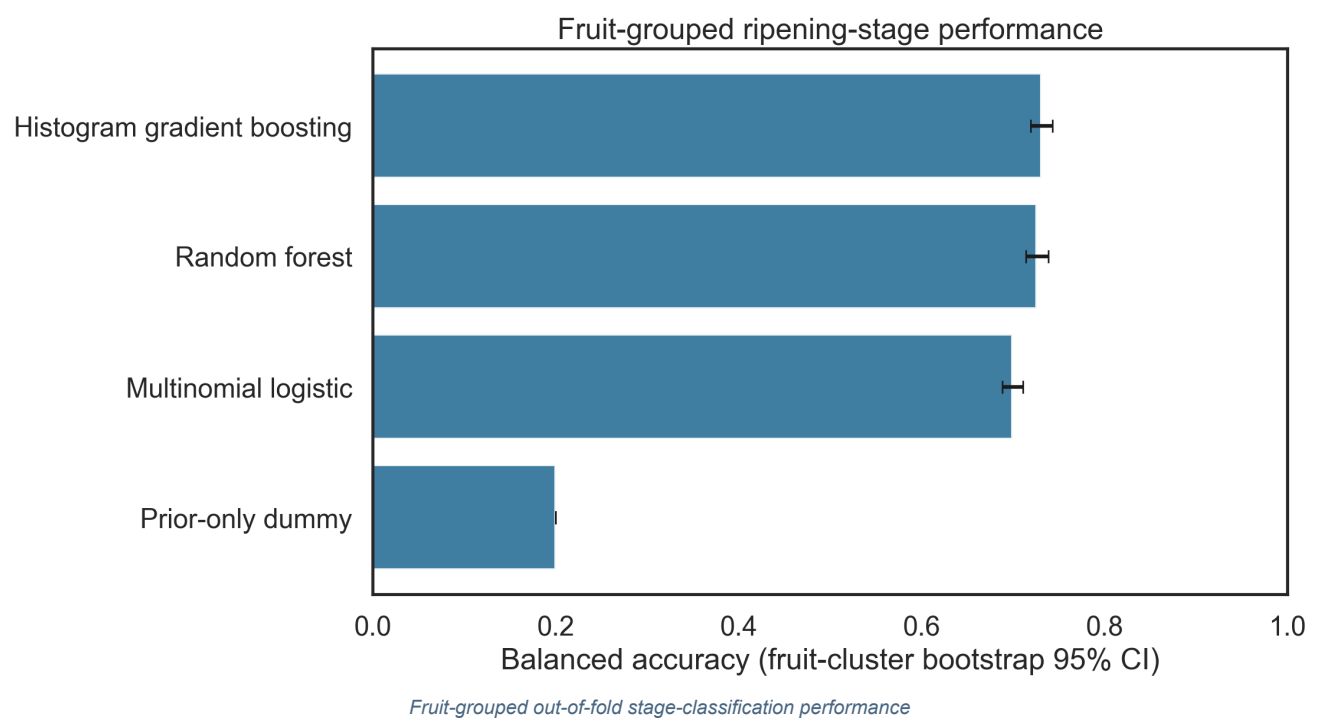
4.3 Reproduced RGB baselines

Foreground-aware colour and compact texture features were extracted without using the filename, fruit ID, day, side, storage condition, or target in the predictor matrix. All records of each fruit stayed in one fold.

DATA-DEMONSTRATED five-fold fruit-grouped results:

- Five-stage histogram-gradient-boosting classifier: accuracy 0.739 (fruit-cluster bootstrap 95% CI 0.727-0.751), balanced accuracy 0.731 (0.719-0.743), macro F1 0.730 (0.719-0.742), ordinal MAE 0.278 (0.264-0.293).
- Retrospective days-to-first-stage-4 model: MAE 1.853 days (1.783-1.922), RMSE 2.636 days (2.542-2.727), R-squared 0.682 (0.655-0.704).
- Entire-storage-group holdout balanced accuracy: 0.497 for T10, 0.688 for T20, and 0.632 for ambient.

The days-to-stage-4 target is derived from the same external composite labels. It is not an independently observed edible-life endpoint. The weak T10 transfer is a strong reason to treat storage conditions and colour-domain shifts as first-class validation problems. Full methods, tables, predictions, split audit, and limitations are in `ml-results.md`.



5. Candidate governing relationships

Every equation below is **PROPOSED** unless explicitly attributed to prior literature. Candidate models are alternatives to compare, not simultaneously accepted truth.

5.1 Temperature-dependent rate constants

An Arrhenius relationship can represent temperature dependence:

$$k_j(T) = A_j \exp\left(-\frac{E_{a,j}}{R_g T_K}\right),$$

where A_j is a pre-exponential factor, $E_{a,j}$ an apparent activation energy, R_g the gas constant, and T_K absolute temperature. A Q10 form may be more identifiable over a narrow range:

$$k_j(T) = k_j(T_{\text{ref}}) Q_{10,j}^{(T - T_{\text{ref}})/10}.$$

Temperature logs must be sufficiently variable and independently replicated to estimate either form. A single chamber per condition confounds condition with chamber.

5.2 Water and mass transfer

A compact fruit-level model is:

$$\frac{dM_w}{dt} = -k_w(T, RH, v) A_f \left(a_w(M_w, T) - RH_f\right),$$

where M_w is water mass, A_f surface area, RH_f ambient fractional RH, v airflow, and a_w an effective surface water activity. Observed total mass also changes through respiration:

$$\frac{dM}{dt} = \frac{dM_w}{dt} - r_C + r_{\text{deposit}},$$

so scale loss must not be equated automatically with water loss.

If spatial moisture gradients matter, a radial/surface PDE candidate is:

$$\frac{\partial W}{\partial t} = \nabla \cdot (D_W(W, T) \nabla W), \quad -D_W \nabla W \cdot \mathbf{n} = h_m(W_s - W_{eq}(T, RH)).$$

This PDE is not identifiable from whole-fruit mass alone. It requires spatial sampling or strongly informative priors and a known geometry.

5.3 Firmness and softening

Simple alternatives include first-order, Weibull, and logistic decay:

$$\frac{dF}{dt} = -k_F(T) F, \quad F(t) = F_{\infty} + (F_0 - F_{\infty}) \exp[-(k_F t)^n],$$

or ethylene-modulated softening:

$$\frac{dF}{dt} = -k_F(T) \frac{E^h}{K_E^h + E^h} (F - F_{\infty}).$$

The last form reflects the type of coupling already explored in avocado literature; its parameters require independent ethylene and standardized firmness measurements. Repeated squeezing is not an acceptable reference because it changes the fruit and lacks traceable force geometry.

5.4 Ethylene and respiration

An autocatalytic ethylene candidate is:

$$\frac{dE}{dt} = k_{E,0}(T) + k_{E,a}(T) \frac{E^h}{K_E^h + E^h} - k_{E,l}(T) E.$$

A dynamic headspace observation model is required:

$$\frac{dC_E}{dt} = \frac{r_{E_{m_f}} V_h - k_{\text{leak}} E (C_E - C_{E, \text{amb}}) - k_{\text{ads}} C_E}{V_h}$$

and similarly for CO₂. For a short, well-mixed, leak-corrected accumulation:

$$r_{\text{CO}_2} \approx \frac{V_h}{m_f} \frac{P}{R_g T_K} \frac{dx_{\text{CO}_2}}{dt}.$$

Sensor response time, chamber volume, fruit displacement, pressure, temperature, leakage, adsorption, and range saturation must be accounted for.

5.5 Pigments and observed colour

$$\frac{dC_{\text{chl}}}{dt} = -k_{\text{chl}}(T) C_{\text{chl}},$$

$$\frac{dA_{\text{anth}}}{dt} = k_A(T) f(R) (A_{\text{max}} - A_{\text{anth}}) - k_{A,d} A_{\text{anth}}.$$

Observed colour is an observation function:

$$y_{\text{colour}} = g(C_{\text{chl}}, A_{\text{anth}}, \text{surface}, \text{illumination}, \text{camera}) + \epsilon,$$

not a direct measurement of either pigment. A chemical pigment subset is needed if pigment concentrations are to be claimed.

5.6 Carbohydrate and concentration balance

$$\frac{dS_{\text{starch}}}{dt} = -k_S(T) S_{\text{starch}},$$

$$\frac{dS_{\text{sol}}}{dt} = Y_{S_k} S_{\text{starch}} - r_{\text{resp, sol}} - r_{\text{other}}.$$

The measured concentration is approximately:

$$C_{\text{sol}}(t) = \frac{S_{\text{sol}}(t)}{M_w(t) + S_{\text{sol}}(t) + M_{\text{other, sol}}(t)}.$$

Therefore Brix can rise through biochemical conversion, water loss, or both. Starch and moisture measurements are mandatory for a starch-to-sugar mass-balance claim.

5.7 Lipid and dry matter

Most avocado lipid deposition occurs before harvest. A conservative postharvest model treats absolute lipid mass as stable or slowly consumed:

$$\frac{dM_L}{dt} = -r_L(T, R), \quad C_{L, \text{fresh}} = \frac{M_L}{M}.$$

Fresh-basis oil percentage can increase as M_w decreases even without lipid synthesis. Direct oil extraction or a validated reference method is needed to calibrate optical estimates.

5.8 Latent ripeness and observation system

A parsimonious latent progression variable can use logistic or Gompertz dynamics:

$$\frac{dR}{dt} = k_R(T, RH, E) R(1-R),$$

or

$$R(t) = \exp[-\exp\{-k(t-t_0)\}].$$

Measured channels are linked through:

$$y_{i,t} = h_i(R_t, M_{w,t}, C_{\text{chl},t}, A_{\text{anth},t}, F_t, E_t, \text{device}_i) + \epsilon_{i,t}.$$

This construction can regularize forecasts, but a one-dimensional R may hide distinct biological pathways and internal disorders. Alternative state dimensions and posterior predictive checks are required.

5.9 Identifiability hierarchy

1. Fit simple descriptive mixed-effects trajectories.
2. Check structural and practical identifiability of each candidate ODE.
3. Add couplings only when corresponding variables are measured.
4. estimate fruit- and batch-level random effects;
5. compare out-of-batch predictive performance and residual structure;
6. reject or simplify models with wide/nonphysical parameters, strong posterior correlations, or failed intervention predictions.

Current public RGB data cannot identify the constants above. Equations, variable definitions, assumptions, and estimation options are collected in Appendix E.

6. Scientific machine learning strategy

6.1 Recommended order

First line: nonlinear mixed-effects ODEs and Bayesian state-space models. They naturally express fruit-to-fruit variation, irregular observations, measurement error, missing destructive states, and uncertainty.

Second line: Gaussian-process or spline state estimation followed by SINDy or constrained symbolic regression. These can propose sparse rate terms but are sensitive to derivative noise, correlated variables, scaling, and unmeasured forcing. Every discovered term must be refitted and tested on held-out batches.

Third line: Neural ODEs or universal differential equations that place a small neural residual beside known mass-balance structure. They are appropriate only after enough independent trajectories exist.

Conditional rather than default: physics-informed neural networks. PINNs are most compelling when a defensible PDE, boundary conditions, and dense spatiotemporal observations exist. Sparse whole-fruit time series do not become mechanistic merely because an ODE residual is added to a loss.

This ordering is informed by the 2026 avocado PINN precedent rather than by an assumption that PINNs are unexplored. The published model demonstrates the feasibility of learning common firmness parameters across literature datasets, while its stated controlled-atmosphere and initial-state limitations reinforce the need for measured storage conditions and multiple biological states ([van der Sman et al.](#)). The proposed method-selection gate also follows the principle in Guo et al. that the balance among knowledge, data volume, prediction goal, efficiency, and interpretability should determine the SciML method ([Guo et al.](#)).

6.2 Candidate methods

Method	Best use here	Advantage	Principal risk
Bayesian state-space model	Latent ripeness/firmness/moisture with noisy sensors	Full uncertainty and missing-state treatment	Prior sensitivity and computational cost
Nonlinear mixed-effects ODE	Population rates plus fruit/batch variation	Interpretable hierarchical kinetics	Structural non-identifiability
Extended/unscented Kalman filter	Online state update on embedded/cloud system	Efficient sequential inference	Gaussian/local-linear assumptions
Particle filter	Nonlinear/non-Gaussian online state	Flexible uncertainty	Degeneracy and computation
Gaussian process	Smooth trajectories and uncertainty	Strong small-data baseline	Scaling and derivative artifacts
SINDy	Sparse candidate equation discovery	Interpretable terms	Noise, hidden variables, unstable derivatives
Symbolic regression	Alternative functional forms	Human-readable candidates	Search overfit and dimensional inconsistency
Neural ODE	Flexible continuous-time latent dynamics	Handles irregular time	Weak mechanistic meaning without constraints
Universal differential equation	Known balance plus learned residual	Separates known/unknown components	Learned residual can absorb confounding
PINN	PDE/ODE-constrained inverse problem	Joint state/parameter fit	Optimization stiffness and false physical confidence

6.3 Scientific validity tests

- enforce units or nondimensionalize;
- require nonnegative states and physically plausible parameter ranges;
- check structural identifiability before numerical fitting;
- use synthetic recovery tests only as pipeline verification, clearly separated from biological evidence;
- compare information criteria, held-out likelihood, forecast error, interval coverage, and residual autocorrelation;
- test intervention predictions across temperature/RH conditions not used for fitting;
- report posterior predictive checks and parameter correlations;
- compare against a simpler empirical baseline;
- avoid claiming an equation is "discovered" unless it survives independent data and targeted perturbation.

7. Conventional machine learning workflow

7.1 Targets

Primary targets:

1. continuous instrumental firmness in N;
2. time to enter a predeclared ready-to-eat window;
3. time to exit the acceptable window or reach an internal-defect endpoint.

Secondary targets:

- moisture and dry matter percentage;
- oil percentage where reference measurements are available;
- ordinal ripening state;
- internal condition/defect class;
- CO₂ production rate;
- calibrated pigment concentrations in a subset.

7.2 Feature sets

• RGB: calibrated L, a, b*, hue, chroma, colour ratios, texture, segmentation masks, and multi-zone variation.

• NIR: dark/white-corrected spectra, absorbance, derivatives, SNV/MSC preprocessing candidates, selected bands, PCA/PLS scores, and spatial variability.

• Environment: temperature/RH histories, vapour-pressure deficit, cumulative thermal exposure, airflow/duty cycle, and storage transitions.

• Mass: absolute and relative change, robust slopes, curvature, and temperature-corrected scale diagnostics.

• Gas: leak/range-corrected CO₂ accumulation slopes, conditioned MOX trajectories, reference ethylene in the validation subset.

• Temporal: time since baseline, change rates, exponentially weighted history, and model-derived latent states.

Every derived feature is computed inside the training partition when it contains fitted parameters. Future time points, endpoint dates, filenames, and post-outcome information are prohibited predictors.

7.3 Models

The progression is:

1. dummy, linear/ridge/logistic, proportional-odds, and PLS/PLS-DA baselines;
2. random forest, gradient boosting, XGBoost or LightGBM, and SVM/SVR;
3. CNN or pretrained visual encoder for sufficiently large image cohorts;
4. temporal convolution, gated recurrent model, or compact transformer only when the number of independent trajectories supports it;
5. early-, intermediate-, and late-fusion multimodal architectures;
6. probabilistic or quantile models for prediction intervals.

Complexity must earn its place on an untouched outer batch. Hyperparameters, feature selection, calibration transfer, and preprocessing are selected inside nested training folds only.

7.4 Validation design

patch/pixel → source image → fruit → batch/harvest → season/orchard/device

All lower-level observations remain inside their parent partition. The minimum reporting set includes:

- nested fruit-grouped cross-validation for development;
- leave-one-batch-out evaluation;
- a locked final harvest/season/device cohort;

- storage-condition stress tests;
- subgroup performance by initial maturity, mass, temperature, RH, and defect;
- complete split manifests and duplicate/near-duplicate checks.

7.5 Metrics and interpretation

Continuous outcomes: MAE, RMSE, R-squared, mean bias, calibration slope, prediction-interval coverage, and error by forecast horizon. Ordinal outcomes: balanced accuracy, macro F1, quadratic-weighted kappa, ordinal MAE, and per-stage recall. Events: time-dependent Brier score, concordance, calibration, and integrated absolute error under interval censoring. Chemometrics: RMSEP, bias, R-squared, RPD, and calibration-transfer error.

Image-level intervals are misleading when one fruit supplies many images. Uncertainty is bootstrapped or modeled at fruit and batch levels.

7.6 Required ablations

- colour only;
- NIR only;
- colour plus NIR;
- optical plus mass/environment;
- optical plus gases;
- all channels;
- current time point only versus longitudinal history;
- discrete LEDs versus full spectrometer;
- empirical model versus empirical plus dynamic state;
- raw device calibration versus transferred calibration.

8. Experimental design

8.1 Staged design

Stage A - metrology and feasibility. Use stable optical targets, check masses, empty chambers, inert volume controls, gas standards, and approximately 24-30 non-study fruits to quantify repeatability, range, drift, heating, repositioning error, carry-over, and scan burden. Biological conclusions are not drawn from this stage.

Stage B - pilot biological study. Estimate variance components, missingness, endpoint frequency, destructive-assay repeatability, and candidate effect sizes. Use at least two independent receipt/harvest batches. The pilot is used to simulate confirmatory power, not to optimize and test the same final model.

Stage C - confirmatory experiment. A planning design contains 360 fruits: three temperatures x two RH levels x three independent harvest batches x 20 fruits. Within each condition-by-batch cell, eight fruits form a repeated non-destructive sentinel cohort and 12 form a sacrifice cohort, with three fruits assigned to each of four prespecified destructive windows. The final sample size and number of destructive windows will be set by blinded variance/power simulation from Stage B.

This design assumes two independently monitored storage chambers or chamber cycles per temperature-RH condition. If condition has only one chamber, the temperature/RH effect is confounded with chamber and must be labeled exploratory. Fruit remains the biological unit; chamber and batch are higher experimental units.

8.2 Fruit inclusion and randomization

Record cultivar verification, supplier/orchard if available, harvest or receipt date, pre-storage, initial mass, external defects, sun/shade side, and initial optical state. Exclude only by preregistered criteria. Within each batch, stratify by initial mass and an initial non-destructive maturity proxy, then randomize fruit to chamber, position, sentinel/sacrifice role, and destructive window. Randomize daily measurement order. Assessors of firmness, internal defects, and chemical assays should be blinded to model predictions and, where possible, storage condition.

8.3 Longitudinal measurement session

1. Verify fruit ID, chamber, scan fixture, sensor clocks, and calibration IDs.
2. Record stable mass after tare with fan and pump off.
3. Capture locked-exposure RGB at indexed sides/azimuths and three axial zones with a white/grey or ColorChecker reference.
4. Acquire a dark signal and reference, then sequential optical measurements at each wavelength or a full NIR spectrum.
5. Place fruit in the headspace chamber; record ambient/empty baseline.
6. Equilibrate for a fixed interval with controlled mixing and LEDs off.
7. Seal for a fixed, range-safe accumulation interval; log CO₂, T, RH, and MOX every 1-5 seconds.
8. Vent or flush until a prespecified recovery criterion is met.
9. Return the fruit promptly to its randomized storage position.
10. Write an immutable session manifest and quality flags.

Daily observation is preferred during rapid 20 C/ambient ripening. Cold-stored fruit may be measured every 24-48 hours after pilot information-loss analysis. Transfers must be short and standardized because measurement itself changes temperature and water loss.

8.4 Destructive reference session

At the assigned window:

- repeat all non-destructive measurements immediately before sacrifice;
- photograph exterior and standardized cut surfaces;
- measure peel-removed puncture/compression firmness at defined sites, probe, speed, depth, and temperature;
- sample matched zones for wet mass and oven-dry or validated moisture analysis;
- calculate dry matter from the same samples;
- measure soluble solids with a calibrated refractometer;
- perform a validated starch/carbohydrate assay if conversion is a claim;
- measure oil by a validated extraction/reference procedure in the designated subset;
- quantify peel chlorophyll and anthocyanin in the designated chemical subset;
- record internal browning, bruising, vascular discoloration, rot, and seed condition using a blinded rubric;
- retain aliquots according to biosafety, chemical, and storage protocols.

8.5 Controls and calibration

Frequency	Required control
Every optical session	Dark, diffuse white, colour target, exposure metadata, saturation flag
Every weighing session	Tare, zero stability, before/after check mass
Every headspace session	Ambient baseline, close/open timestamps, recovery criterion
Daily	Empty chamber or scheduled blank, leak/recovery control, clock sync, cleaning record
Weekly/block	CO2 zero/span, multi-point mass calibration, colour-target condition, reference-object repeatability
Campaign boundaries	Wavelength/photometric validation, gas calibration, inter-device comparison

Acceptance criteria must be numerical and locked after metrology, including repeat-scan coefficient of variation, repositioning error, maximum detector saturation, scale creep, chamber leakage, gas range compliance, and blank recovery.

8.6 Ground-truth endpoint definition

"Ripe," "ready," and "expired" are operational endpoints, not universal natural constants. Before confirmatory collection, a panel consisting of the biological lead, postharvest scientist, statistician, and intended user must define:

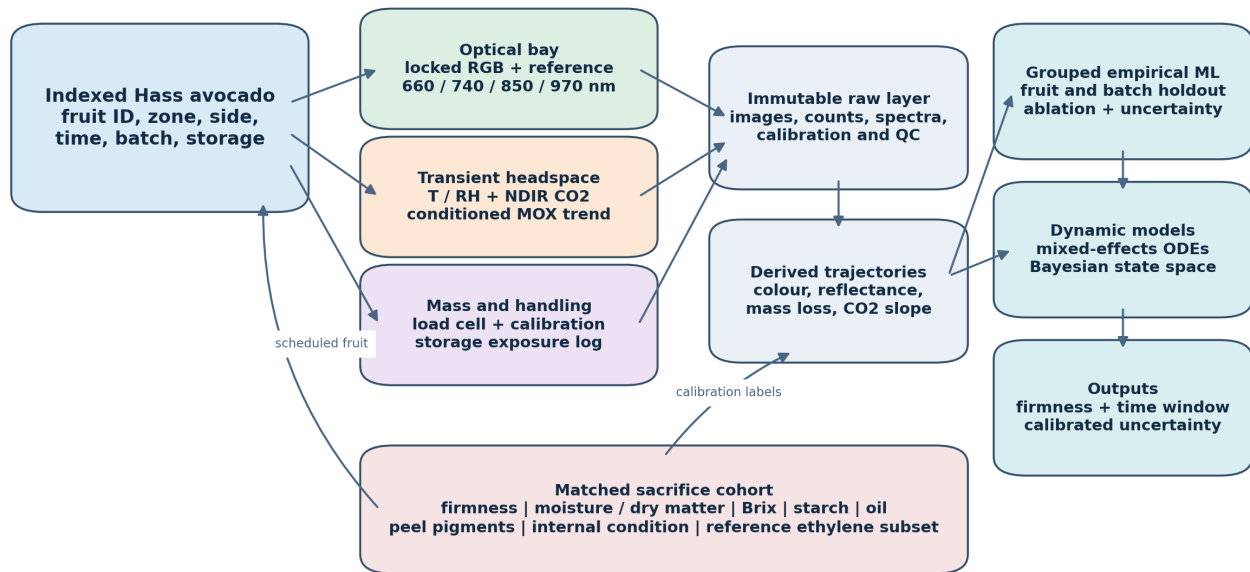
- the instrumental firmness window;
- acceptable internal-condition rubric;
- whether flavour/sensory information is required;
- the event used for remaining useful life;
- handling of fruits that never reach an endpoint during follow-up;
- interval-censoring rules;
- thresholds for actionable error.

The endpoint cannot be redefined after viewing locked test performance.

9. Prototype and materials to procure

Proposed multimodal avocado characterization system

Repeated non-destructive measurements linked to matched destructive ground truth



Claims boundary: RGB, NIR, mass, CO2, and MOX are observations. Internal chemistry is a calibrated prediction only when matched reference assays support it.

Overview of the proposed multimodal system and evidence flow

9.1 MVP

The minimum system uses:

- Espressif ESP32-S3-DevKitC-1-N8R8 for control/logging;
- Raspberry Pi Zero 2 W (SC1176) and Camera Module 3 standard;
- Calibrite ColorChecker Classic Mini (CCC-MINI);
- Sensirion SHT45 and SCD30;
- Bosch BME688 used only as a conditioned broad MOX gas channel;
- SparkFun TAL220 10-kg load cell and HX711;
- Adafruit ADS1115 and microSD interface;
- Vishay BPW34 silicon photodiode;
- current-regulated 660/740/850/970-nm LED set;
- light-isolated enclosure, indexed cradle, fan, power, analog front end,

tubing, calibration masses, baffles, and wiring.

The MVP directly measures images, broad optical responses, chamber T/RH, mass, CO2 concentration, and MOX resistance. It does not directly measure internal water, oil, starch, ethanol, or reference ethylene.

The silicon-range choice is supported as a cost-reduction research direction by the 400-1100 nm analysis of Melado-Herreros et al.; it is not a claim that four LEDs reproduce their 380-2000 nm spectrometer ([Melado-Herreros et al.](#)). Because Mishra et al. demonstrated degraded Vis-NIR firmness transfer under peel dehydration, the MVP must log RH and mass and explicitly test dehydration-condition holdouts ([Mishra et al.](#)).

9.2 Research configuration

The research baseline adds a 900-1700 nm InGaAs spectrometer, stabilized source, interactance/reflectance probe, certified diffuse reference, inert headspace vessel and PTFE/PFA flow path, reference ethylene analyzer or validated GC method, digital refractometer, standardized penetrometer or texture analyzer, moisture/dry-matter equipment, and chemical assay resources. Extending to 2500 nm is justified only when 1720/2300-nm lipid-sensitive information is an explicit objective.

9.3 Budget and procurement gate

Tier	Audited planning amount	Interpretation
MVP priced core	US\$443.67	Current/listed components with documented price status
MVP allowances	US\$279.95	Fabrication, optics, tubing, electronics, calibration allowances
MVP total	US\$723.62	Procurement planning figure, excluding taxes/shipping
Research additions assuming shared core instruments	US\$6,591.51	Purchased components and allowances
Buy-new major-instrument proxy	US\$43,466.01	Non-quotation; includes historical proxy and current listed instruments
Optional Hamamatsu C11708MA	EUR650.53 ex VAT	Mid-tier 640-1050-nm option, not a SWIR replacement

Prices are time-sensitive. Re-quote before ordering. The exact 44-row BOM, source ledger, limitations, alternatives, and the two locally retained license-verified product photographs are in Appendix D and procurement/.

10. Mechanical design and data architecture

The apparatus has two coupled bays:

1. a light-isolated optical bay with fixed camera/fruit/LED/detector geometry, indexed fruit orientation, matte baffles, and removable dark/white references;
2. a temporarily sealable, well-mixed headspace bay with controlled accumulation, environmental sensing, range checks, and forced recovery.

Optical illumination should be 30-45 degrees from surface normal to reduce specular reflection, with the camera approximately normal to the peel. Scan the stem shoulder, equator, and blossom shoulder at multiple indexed azimuths. Camera exposure, white balance, focus, distance, LED current, integration time, detector gain, and temperature must be recorded.

The chamber is not long-term storage. Permanent sealing risks hypoxia, condensation, microbial changes, and history-dependent gas buildup. A flush-equilibrate-accumulate-vent cycle separates respiration observation from storage.

The raw data contract uses:

```
study_id / batch_id / fruit_id / session_id / timestamp_utc /
storage_condition / chamber_id / scan_zone / azimuth / device_id /
firmware_version / calibration_id / raw_value / unit / quality_flag
```

Original images and sensor counts are immutable. Calibrated and derived values are written to new layers with code and calibration version. Every destructive sample remains linked to its complete non-destructive history.

Detailed optical, gas-flow, electronics, calibration, and acceptance diagrams appear in `prototype-architecture.md`.

11. Variables and measurement rationale

11.1 Independent and assigned variables

Variable	Unit/domain	Role
Storage temperature	degrees C	Assigned factor; continuous log also retained
Relative humidity	% RH	Assigned factor; continuous log also retained
Airflow/mixing duty	defined setting	Controlled factor/covariate
Harvest/receipt batch	categorical	Independent block/domain
Storage time	hours/days	Longitudinal coordinate
Chamber/device	identifier	Technical random effect/domain

11.2 Primary dependent variables

Variable	Method	Status
Firmness	Standardized destructive force protocol	Direct reference
Time into/out of acceptable window	Prespecified composite, interval-censored	Derived primary endpoint
Internal defect/acceptability	Blinded cut-surface rubric	Direct ordinal/binary reference

11.3 Non-destructive observations

Channel	Directly observed	Defensible derived quantity
RGB	Pixel values under calibration	Lab*, hue, chroma, texture, rates
NIR	Reference-corrected intensity/spectrum	Calibrated latent/chemical predictions
Scale	Force/mass	Net mass trajectory
SHT45	Air T/RH	Exposure history, VPD
NDIR CO2	Chamber CO2	Corrected accumulation slope
MOX	Gas resistance plus T/RH	Conditioned nonspecific VOC feature
Low-cost C2H4 bridge	Cross-sensitive indicated current	Exploratory trend after reference comparison

11.4 Destructive/analytical variables

Moisture, dry matter, oil, Brix, starch/carbohydrate, peel pigments, firmness, internal defect, and reference ethylene require destructive or laboratory-grade methods for the claims proposed. A low-cost optical estimate remains a model output whose validity depends on these reference labels.

11.5 Confounders and nuisance variables

Cultivar/subclone, orchard, flowering/harvest maturity, pre-cooling, cold storage duration, supplier handling, sun exposure, fruit size and geometry, skin damage, bloom, bruising, disease, chamber, device, operator, calibration lot, scan position, condensate, and fruit-to-fruit biological variation are recorded or controlled. Unrecorded prehistory is acknowledged rather than absorbed into a universal ripeness claim.

12. Statistical analysis plan

12.1 Analysis populations

- metrology population: stable controls and all calibration trials;
- intention-to-observe biological population: every randomized fruit;
- valid-sensor population: records passing prespecified technical QC;

- destructive-reference population: all assigned sacrifices, including failed

assays with reason codes;

- locked external population: untouched batch/season/device cohort.

Exclusions are reported by fruit and reason. Failed measurements are not silently deleted.

12.2 Descriptive and experimental analyses

For repeated continuous outcomes:

$$y_{\{ijkt\}} = \beta_0 + \beta_{TT_i} + \beta_{HH_j} + \beta_{TH} T_{iH_j} + f(t) + \text{interactions} + b_{\{\mathrm{batch}\}} + b_{\{\mathrm{chamber}\}} + b_{\{\mathrm{fruit}\}} + \epsilon.$$

Use flexible time effects or candidate kinetics, fruit random intercepts/slopes, batch and chamber effects, and heteroscedastic/autocorrelated residuals where supported. Destructive outcomes are analyzed with condition, time, batch, and chamber hierarchy but no repeated fruit effect.

Time-to-ready/end-of-quality is analyzed as interval-censored survival, with frailty/random effects for batch and chamber. Competing internal-defect events are modeled or reported separately.

12.3 Sample size

The 360-fruit confirmatory layout is a planning design, not a claim of achieved power. Final n will be selected by simulation using pilot estimates for:

- batch and chamber intraclass correlation;
- fruit-level residual variance and serial correlation;
- endpoint censoring and missingness;
- smallest useful difference in firmness MAE and remaining-time MAE;
- number of outer batches available;
- multiplicity across primary endpoints.

The effective sample size is driven by independent fruits, batches, and chambers - never by images, pixels, or spectra.

12.4 Missingness and censoring

Record reason codes for sensor failure, saturation, absent image, assay failure, fruit damage, contamination, and endpoint not observed. Mixed/state-space models can use incomplete trajectories under explicit assumptions. Sensitivity analyses include complete-case, multiple-imputation or model-based missingness, and worst-reasonable scenarios for informative fruit loss. Fruits not reaching the endpoint are censored, not assigned an arbitrary remaining time.

12.5 Multiplicity and uncertainty

Define a small primary family before collection. Use hierarchical testing or false-discovery-rate control for secondary sensor/biomarker screens. Report effect sizes and 95% intervals, not only p-values. Bootstrap at fruit and batch levels, propagate calibration uncertainty, and present prediction intervals with observed coverage.

12.6 Model comparison

All competing models use identical outer folds and target definitions. Compare paired fruit/batch errors, calibration, coverage, subgroup behavior, and computation. Statistical superiority without a meaningful improvement or transfer robustness is insufficient for adoption.

13. Project timeline and phase execution

13.1 Work already executed in this workspace

Phase	Completed output
1	Public-source discovery, licensing decisions, dataset inventory
2	Archive inspection, schema reconciliation, data dictionary, QA
3	Fruit-level EDA and publication-ready figures
4	Fruit-grouped classification/regression baselines and storage holdout
5	Candidate governing relationships and identifiability plan
6	Primary/secondary questions, hypotheses, nulls
7	Matched longitudinal/sacrifice experimental design
8	Measurement matrix and rationale
9	MVP and research-grade prototype architecture
10	44-row sourced procurement catalogue
11	Statistical analysis and validation plan
12	Integrated proposal
13	Draft communication to prospective collaborators
14	Appendices, source/QA ledgers, scripts, figures, and rendered PDF

13.2 Future experimental timeline

Month	Activity	Gate
1-2	Supervisor/stakeholder endpoint definition, protocol review, data-management plan	Signed endpoint and claims matrix
2-3	Re-quote/procure MVP, fabricate chamber and fixture	BOM and safety review
3-4	Firmware, logging schema, camera and optical integration	Schema validation
4-5	Metrology: optical repeatability, mass drift, chamber leaks, gas range, carry-over	Numerical acceptance criteria pass
5-6	Pilot fruit cohort and destructive-method training	Pilot QA report
6-7	Blinded variance/power simulation; preregistration; model freeze plan	Confirmatory protocol approval
7-10	Multi-batch confirmatory collection	Batch-complete QA, no outcome-driven retraining
8-11	Reference assays, data lock, conventional ML and statistics	Locked external set untouched
10-12	Dynamic/scientific-ML comparison and identifiability analysis	Posterior/holdout diagnostics
12-13	External-device or additional-season validation	Deployment boundary decision
13-15	Manuscript, dataset/code release subject to rights, prototype revision	Reproducibility audit

14. Risk analysis and mitigation

Risk	Consequence	Mitigation / decision rule
Same fruit appears across splits	Inflated performance	Immutable fruit IDs and automated zero-overlap split audit
One chamber per condition	Chamber-condition confounding	Independent chamber replication or label effects exploratory
Colour/internal state desynchronization	Unsafe ripe/ready inference	Multimodal sensing and internal/firmness reference
NIR scattering/water/lipid confounding	False chemical specificity	Broad-spectrum reference calibration, geometry controls, non-claims
MOX/electrochemical cross-sensitivity	Mislabelled ethylene/VOC result	Reference comparison, challenge gases, broad-feature labeling
CO ₂ range saturation or leaks	Biased respiration slope	Shorter accumulation, larger chamber, leak/blank correction, flags
Optical heating and condensation	Alters fruit/sensor signal	Separate phases, thermal logs, controlled duty, recovery
Scale drift or contact forces	False mass-loss trend	Isolated cradle, calibration/creep checks, fan off
Destructive assay variability	Weak labels	Frozen protocol, operator training, duplicates, blinded QC
Dataset shift across batch/season/device	Deployment failure	Locked external cohorts and calibration-transfer study
Endpoint defined after results	Optimistic bias	Stakeholder-signed preregistration
Too few independent batches	Unstable generalization estimate	Prioritize batches over extra views; hierarchical uncertainty
Attrition/non-ripening fruit	Biased remaining-time model	Interval censoring and missingness sensitivity
Overparameterized ODE	Non-identifiable rates	Structural checks, staged complexity, priors, intervention validation
Public-data rights ambiguity	Legal/reproducibility risk	Acquire only verified licenses; request written permission
Price/availability change	Prototype delay	Re-quote, approved alternatives, staged purchasing
3D-print/enclosure outgassing	Headspace artifacts	Blank/recovery study; inert materials for analytical configuration
Chemical/solvent work	Safety risk	Institutional SOPs, training, fume hood, waste process

Stop/go decisions are evidence based. For example, if discrete 970-nm repeatability is poor or adds no outer-fold information, it remains a teaching channel rather than being marketed as moisture sensing. If a low-cost ethylene bridge fails analytical agreement, it may remain a nonspecific feature but cannot retain an ethylene-concentration label.

15. Expected contributions, limitations, and future work

15.1 Expected contributions

1. A public-data audit that distinguishes images from biological sample size and documents access/licensing boundaries.
2. Reproducible fruit-grouped RGB baselines and a quantified storage-distribution shift.
3. A procurement-ready, claims-aware multimodal prototype.
4. A matched longitudinal/destructive protocol capable of calibrating optical and gas features to real reference quantities.
5. A common validation framework for black-box, statistical, kinetic, and scientific-ML models.
6. An explicit identifiability and non-claims framework.

15.2 Limitations that remain even after the proposed study

- One cultivar and limited sources do not establish general avocado validity.
- Ready-to-eat quality is partly sensory and use-case dependent.

- Destructive matched cohorts approximate, rather than reveal, the unobserved internal state of a surviving sentinel fruit.

- A chamber measurement is not the same as a supermarket or home environment.
- Spectrometer calibration can be device- and geometry-specific.
- Mechanistic models are simplified representations of coupled biology.
- A statistically accurate forecast may still be operationally unhelpful if its uncertainty is too wide or its workflow too burdensome.

15.3 Future work after successful external validation

- edge inference with model cards and abstention under domain shift;
- smartphone or kiosk integration with a calibration accessory;
- industrial sorting and robotic handling;
- harvesting decisions based on orchard maturity rather than only postharvest ripeness;
- other cultivars and fruit species with separate calibration;
- cloud longitudinal database with consent, governance, and drift monitoring;
- transfer learning and calibration transfer between devices;
- federated learning only where multiple institutions have comparable protocols and governance;
- digital-twin forecasting constrained by measured mass, respiration, and temperature histories;
- intervention optimization for storage temperature, RH, and venting;
- consumer studies testing whether predictions actually reduce waste.

Commercial deployment, phone-only claims, and automated decisions are future possibilities, not deliverables of the initial experiment.

16. Deliverables and decision request

The complete package contains:

- source PDFs and evidence-corrected paper summaries;
- licensed public datasets and immutable checksums;
- dataset inventory, QA report, processed tables, and data dictionary;
- EDA scripts, figures, and machine-readable outputs;
- grouped ML training, out-of-fold predictions, split audit, metrics, and model;
- candidate mathematical models and scientific-ML plan;
- experimental protocol, measurement rationale, and statistical plan;
- prototype diagrams, 44-row procurement BOM, source ledger, exact MPNs, and licensed component photographs;
- a message draft for prospective collaborators;
- reproducibility scripts, repository structure, appendices, and rendered PDF.

The immediate decision requested from collaborators is whether to proceed to a metrology-first MVP build and a two-batch pilot, and which laboratory resources are available for NIR, reference ethylene, firmness, moisture/dry matter, oil, pigment, and carbohydrate assays. Those answers determine whether the first study can support internal-chemistry claims or should remain a rigorously bounded optical/environmental feasibility study.

Appendix map

- Appendix A - Existing papers
- Appendix B - Verified bibliography
- Appendix C - Public datasets
- Appendix D - Hardware catalogue
- Appendix E - Candidate mathematical models
- Appendix F - Software stack
- Appendix G - Repository structure
- Appendix H - Data quality and data dictionary
- Appendix I - Machine-learning results
- Appendix J - Statistical analysis plan
- Appendix K - Reproducibility manifest
- Appendix L - Risks, claims, and evidence matrix

Appendix A - Existing Papers Supplied for Review

Evidence-audit date: 30 July 2026

Evidence labels

- **[V-P] Primary-source verified:** verified against a publisher, institutional, author-hosted, or repository primary source.
- **[V-L] Local-full-text verified:** verified directly in the supplied local PDF.
- **[A] Audit assessment:** an interpretation or methodological assessment based on the reported design; it is not a claim made verbatim by the authors.
- **[U] Unresolved:** the available source did not establish the point, or an external permission/metadata check remains necessary.

Reported model performance is reproduced as a claim of the cited paper, not independently replicated performance.

A.1 RipeTrack: Assessing Fruit Ripeness and Remaining Lifetime Using Smartphones

Full citation

Waseem, M. S., Sharma, N., & Hefeeda, M. (2026). RipeTrack: Assessing Fruit Ripeness and Remaining Lifetime Using Smartphones. *IEEE Transactions on Mobile Computing*, 25(1), 830-845. <https://doi.org/10.1109/TMC.2025.3599917>.
[V-P, V-L]

Primary access:

- [Institutional publication record](#)
- [Author-hosted manuscript](#)
- [RipeTrack code and data-link repository](#)

The article appears in the 2026 volume; the 2025 embedded in the DOI is not the publication year. [V-P]

Research objective

RipeTrack addresses two related tasks:

1. classification of fruit as unripe, ripe, or expired; and
2. estimation of the fruit’s remaining lifetime.

The proposed consumer-facing system uses smartphone sensing and machine-learning models to infer optical changes associated with ripening without cutting the fruit. Its intended value is scalable, non-invasive assessment for retailers and consumers rather than laboratory chemical analysis. [V-P, V-L]

Experimental and technical approach

The authors first collected hyperspectral images and ethylene measurements across the observed postharvest lives of several fruit types. The laboratory hyperspectral system used a Specim IQ camera covering approximately 400-1000 nm with 204 bands and approximately 3 nm spectral sampling. Ethylene was measured using an FD-90A-C2H4 detector specified in the paper as covering 0-100 ppm with 1 ppm accuracy. Mobile experiments used Google Pixel 4XL and OnePlus 8 Pro phones. [V-L]

The full laboratory dataset reported in Table 1 of the paper is:

Fruit/cultivar	Independent fruits	Hyperspectral images	Ethylene readings	Observed lifetime
Organic avocado	10	460	230	11 days
Hass avocado	3	234	117	19 days

Fruit/cultivar	Independent fruits	Hyperspectral images	Ethylene readings	Observed lifetime
Bartlett pear	11	382	209	12 days
Bosc pear	3	276	138	40 days
Banana	12	279	168	7 days
Nectarine	3	138	138	16 days
Mango	6	144	144	16 days
Total	48	1,913	1,144	-

Evidence: counts transcribed from the supplied paper. [V-L]

The mobile dataset contains 3,865 paired RGB and near-infrared captures: 2,695 from the Pixel 4XL and 1,170 from the OnePlus 8 Pro. The paper reconstructs a richer spectral representation from phone measurements and then uses learned models for ripeness and remaining-life prediction. The authors also examine transfer learning to additional fruits and measurements under varied illumination. [V-L]

Main reported findings

- Avocado ripeness classification accuracy: **95%**.
- Pear ripeness classification accuracy: **98%**.
- Avocado remaining-lifetime prediction accuracy: **93%**.
- Pear remaining-lifetime prediction accuracy: **97%**.
- The paper reports that the mobile ripeness test set used fruits not observed during training.
- The paper reports extension to bananas, mangoes, and nectarines through transfer learning and tests in home and grocery-store illumination.

The 93% and 97% figures are accuracies for an 11-class remaining-life-percentage target (0%, 10%, ..., 100%). The target is derived from days until the start of RipeTrack's ethylene-defined *Expired* stage, with approximately one- to two-day granularity depending on fruit lifetime. These percentages are not continuous regression accuracy, MAE in days, or validation against a sensory usable-life endpoint. [V-L, A]

These are the authors' reported results and require independent replication before they are treated as expected prototype performance. [V-P, V-L]

Contribution to the proposed research

RipeTrack supplies a strong systems precedent for:

- multimodal visible and near-infrared sensing;
- moving from laboratory hyperspectral evidence toward consumer hardware;
- reporting an intuitive ripeness state and remaining-life output;
- testing multiple phones and lighting conditions; and
- using transfer learning for additional fruit types.

The proposed avocado study extends this direction by adding explicit biological ground truth-firmness, mass loss, moisture/dry matter, soluble solids, starch, oil, pigment, and potentially gas measurements-and by fitting interpretable longitudinal state and rate models rather than treating each image as an independent black-box sample. [A]

Limitations and cautions

1. **The biological sample size is much smaller than the image count.** The avocado results derive from 13 independent fruits in total, including only three Hass fruits. Repeated images increase temporal resolution but do not create new independent biological replicates. [V-L, A]

2. **Cultivar and supply-chain generalization remain uncertain.** Organic and Hass avocado groups are small and do not establish robustness across orchards, harvest maturity, season, cold-chain history, or country of origin. [A]
3. **The reconstruction split needs a grouping audit.** A 70/15/15 split is reported for hyperspectral reconstruction, but the manuscript does not clearly establish that all repeated views and days from a fruit were confined to one partition. Any re-analysis must split by fruit identity. [V-L, A]
4. **Accuracy is not a complete shelf-life metric.** Future evaluation should report class balance, confusion matrices, ordinal error, MAE/RMSE in days, calibration intervals, and performance on wholly independent harvest batches. [A]
5. **Optical features are not uniquely attributable to water.** Visible/NIR changes may combine water absorption, lipid- and carbohydrate-related absorption, pigment changes, geometry, and scattering from softening tissue. Wavelength interpretations require destructive calibration. [A]
6. **Dataset reuse rights are unresolved.** The public GitHub repository links datasets hosted on Google Drive, but no explicit dataset licence was located during this audit. Public downloadability alone does not grant reuse or redistribution rights. Written clarification is required before incorporating the data into a redistributed project. [U]

Bottom-line interpretation

RipeTrack demonstrates the feasibility of a phone-oriented optical pipeline and provides a valuable reference architecture. It does **not** by itself establish a universally valid avocado ripeness model, nor does it identify a single causal chemical mechanism behind the measured spectra. The strongest extension is therefore a fruit-grouped, batch-independent, multimodal longitudinal experiment with destructive chemical and mechanical calibration. [A]

A.2 Non-destructive Methods for Fruit Quality Evaluation

Full citation

Bratu, A.-M., Popa, C., Bojan, M., Logofatu, P. C., & Petrus, M. (2021). Non-destructive methods for fruit quality evaluation. *Scientific Reports*, 11, 7782. <https://doi.org/10.1038/s41598-021-87530-2>. [V-P, V-L]

Primary access:

- [Scientific Reports article](#)
- [Article PDF](#)

The article is open access under CC BY 4.0. This licence covers the article and included material unless a credit line says otherwise; it does not establish the existence or reuse licence of a separate raw dataset. [V-P]

Research objective

This study evaluates the deterioration of stored apples by combining:

1. laser photoacoustic spectroscopy (LPAS) for ethylene and ethanol in the fruit's emitted gases; and
2. multispectral imaging using visible spectral filters for external reflectance and defect-related changes.

The work concerns apples, not avocados, and its quantitative gas trajectories should not be transferred directly to avocado. [V-P, V-L, A]

Experimental approach

Twenty apples-five Golden Delicious, five Gala, five Granny Smith, and five Starkrimson-were monitored at room temperature over 35 days. Measurements were taken on days 1, 7, 14, 21, 28, and 35. For gas analysis, each fruit was placed in a 150 cm³ glass cuvette, flushed with synthetic air, and connected to a CO₂-laser photoacoustic system. The study selected a laser line for ethylene and another for ethanol. External measurements used an imaging system with eight visible-region spectral filters and tracked histograms from selected surface regions. [V-P, V-L]

Main supported findings

- Ethylene emission decreased over storage for all four cultivars, with a pronounced decline around the second and third weeks.
- Ethanol was initially low and increased during storage, becoming especially elevated once fruit showed advanced deterioration and fermentation.
- Cultivar-specific trajectories were evident: Golden and Gala began with higher ethylene, while Starkrimson showed comparatively high ethanol.
- Visible defects, wrinkling, brown patches, and mass loss became apparent over time.

These findings support gas sensing as a potentially informative longitudinal modality, but the study's five fruits per cultivar provide limited statistical power and do not establish an avocado-specific gas law. [V-P, V-L, A]

Required correction to an overstrong reading of the imaging results

The multispectral imaging results were **not** a strong demonstration that image changes tracked gas deterioration. The authors explicitly report that practical impediments prevented uniform experimental conditions and that they were mostly unsuccessful in finding clear temporal patterns. In the discussion, they state that the histogram and global-plot changes generally could not be separated from statistical or systematic error. A slight overall shift toward darker tones was found for the more extensively observed Granny Smith case. [V-P, V-L]

Accordingly, the defensible interpretation is:

The paper demonstrates a precise gas-measurement approach and an exploratory attempt to combine it with multispectral imaging; the imaging evidence was mostly inconclusive rather than a validated external deterioration model.

[A]

Contribution to the proposed research

The paper remains relevant as a methodological precedent for:

- pairing an external optical signal with an internal/volatile signal;
- measuring ethylene and ethanol longitudinally;
- accounting for cultivar-dependent gas behavior; and
- recognizing the calibration and repeatability demands of spectral imaging.

For the avocado proposal, gas sensing should be treated as an experimentally calibrated complementary modality. Low-cost VOC sensors are not automatically selective for ethylene or ethanol, and any chamber measurement must control fruit mass, chamber volume, accumulation time, flow, temperature, humidity, background gas, sensor cross-sensitivity, and sensor drift. [A]

Limitations and cautions

1. **Species mismatch:** apple physiology and peel gas diffusion cannot supply avocado-specific parameters without new experiments. [A]
2. **Small biological sample:** five apples per cultivar with repeated measurements; repeated time points are not independent replicates. [V-P, A]
3. **Limited imaging evidence:** most spectral-histogram changes were not distinguishable from experimental variability. [V-P, V-L]
4. **Descriptive analysis:** the paper does not provide a validated remaining-life model, independent test cohort, or modern grouped machine-learning evaluation. [A]
5. **No public raw-data repository located:** the article is open access, but this audit found no linked machine-readable release of the gas time series and multispectral images. [U]

Bottom-line interpretation

The paper supports the concept of multimodal gas-plus-optical monitoring but does not validate a deployable imaging predictor. Its most useful lesson for the proposed avocado work is experimental: controlled chambers, selective/calibrated gas measurements, biological replication, and synchronized optical measurements are necessary before coupled kinetic or machine-learning models can be trusted. [A]

A.3 How the two supplied papers jointly frame the proposal

Dimension	RipeTrack	Bratu et al.	Proposed avocado extension
Target setting	Smartphone/consumer and retail	Laboratory/storage research	Calibrated prototype progressing toward deployable hardware
Fruit scope	Avocado, pear, banana, mango, nectarine	Four apple cultivars	Hass avocado with documented harvest and storage batches
Primary modalities	RGB, near-infrared, reconstructed hyperspectral signatures, ethylene	LPAS ethylene/ethanol and exploratory multispectral imaging	RGB/colour, NIR, firmness, fruit mass, environment, selective gases, destructive chemistry
Primary outputs	Ripeness class and remaining lifetime	Longitudinal gas and image changes	Ripeness state, firmness, moisture/dry matter, days to edible/overripe thresholds, uncertainty
Key strength	Practical sensing architecture	Precise trace-gas methodology	Multimodal biological calibration and interpretable dynamics
Key limitation	Very small independent avocado sample, especially Hass	Apple-only, five fruits/cultivar, weak imaging result	Must be addressed through fruit-grouped, batch-independent design

Synthesis [A]: RipeTrack shows how optical sensing can become an accessible prediction system; Bratu et al. shows how internal volatile measurements can be collected alongside imaging. The proposed work should combine the practical orientation of the former with the controlled multimodal measurement discipline of the latter, while avoiding image-level leakage, unsupported chemical attribution, and overstatement of exploratory results.

Appendix B - Evidence-Audited Bibliography

Verification date: 30 July 2026

Evidence labels

- **[V-P]** Bibliographic metadata verified against a publisher, DOI, institutional, or repository primary source.
- **[V-L]** Methods or results also checked in a locally supplied full-text PDF.
- **[R]** Repository or dataset record; access and licensing must be read independently of the associated article.
- **[U]** A reuse licence, raw-data release, or another stated field remains unresolved.

DOI links are used as persistent primary routes. A paper's open-access licence does not automatically license a separately hosted raw dataset, and the existence of a public repository does not by itself grant reuse or redistribution rights.

B.1 Supplied core papers

1. Waseem, M. S., Sharma, N., & Hefeeda, M. (2026). RipeTrack: Assessing Fruit Ripeness and Remaining Lifetime Using Smartphones. *IEEE Transactions on Mobile Computing*, 25(1), 830-845. DOI: [10.1109/TMC.2025.3599917](https://doi.org/10.1109/TMC.2025.3599917). [Institutional record](#). [Author manuscript](#). **[V-P, V-L]**
2. Bratu, A.-M., Popa, C., Bojan, M., Logofatu, P. C., & Petrus, M. (2021). Non-destructive methods for fruit quality evaluation. *Scientific Reports*, 11, 7782. DOI: [10.1038/s41598-021-87530-2](https://doi.org/10.1038/s41598-021-87530-2). [Publisher article](#). **[V-P, V-L]**

B.2 Public-dataset and machine-learning studies

3. Xavier, P., Rodrigues, P. M., & Silva, C. L. M. (2024). Shelf-Life Management and Ripening Assessment of 'Hass' Avocado (*Persea americana*) Using Deep Learning Approaches. *Foods*, 13(8), 1150. DOI: [10.3390/foods13081150](https://doi.org/10.3390/foods13081150). [Full text](#). **[V-P]**
4. Varga, L. A., Makowski, J., & Zell, A. (2021). Measuring the Ripeness of Fruit with Hyperspectral Imaging and Deep Learning. In *2021 International Joint Conference on Neural Networks (IJCNN)* (pp. 1-8). IEEE. DOI: [10.1109/IJCNN52387.2021.9533728](https://doi.org/10.1109/IJCNN52387.2021.9533728). [Author preprint](#). **[V-P]**
5. Han, Y., Bai, S. H., Trueman, S. J., Khoshelham, K., & Kämper, W. (2023). Predicting the ripening time of 'Hass' and 'Shepard' avocado fruit by hyperspectral imaging. *Precision Agriculture*, 24, 1889-1905. DOI: [10.1007/s11119-023-10022-y](https://doi.org/10.1007/s11119-023-10022-y). [Publisher article](#). **[V-P]**
6. Davur, Y. J., Kämper, W., Khoshelham, K., Trueman, S. J., & Hosseini Bai, S. (2023). Estimating the Ripeness of Hass Avocado Fruit Using Deep Learning with Hyperspectral Imaging. *Horticulturae*, 9(5), 599. DOI: [10.3390/horticulturae9050599](https://doi.org/10.3390/horticulturae9050599). [Publisher article](#). **[V-P, V-L]**
7. Lee, I.-H., Li, Z., & Ma, L. (2025). Explainable AI and mobile imaging for non-destructive avocado ripeness and internal quality assessment to reduce food waste. *Current Research in Food Science*, 11, 101196. DOI: [10.1016/j.crfs.2025.101196](https://doi.org/10.1016/j.crfs.2025.101196). [Publisher article](#). [PubMed record](#). **[V-P]**
8. Pinto, J., Rueda-Chacón, H., & Arguello, H. (2019). Classification of Hass avocado (*Persea americana* Mill.) in terms of its ripening via hyperspectral images [Clasificación de aguacate Hass (*Persea americana* Mill.) en términos de su madurez por medio de imágenes hiperespectrales]. *Tecnológicas*, 22(45), 109-128. DOI: [10.22430/22565337.1232](https://doi.org/10.22430/22565337.1232). [Publisher article](#). **[V-P]**
9. Cho, B.-H., Koyama, K., & Koseki, S. (2021). Determination of 'Hass' avocado ripeness during storage by a smartphone camera using artificial neural network and support vector regression. *Journal of Food Measurement and Characterization*, 15(2), 2021-2030. DOI: [10.1007/s11694-020-00793-7](https://doi.org/10.1007/s11694-020-00793-7). [Publisher article](#). **[V-P]**

B.3 Spectroscopy, maturity, water, and chemical ground truth

10. Clark, C. J., McGlone, V. A., Requejo, C., White, A., & Woolf, A. B. (2003). Dry matter determination in 'Hass' avocado by NIR spectroscopy. *Postharvest Biology and Technology*, 29(3), 301-308. DOI: [10.1016/S0925-5214\(03\)00046-2](https://doi.org/10.1016/S0925-5214(03)00046-2). [V-P]
11. Olarewaju, O. O., Bertling, I., & Magwaza, L. S. (2016). Non-destructive evaluation of avocado fruit maturity using near infrared spectroscopy and PLS regression models. *Scientia Horticulturae*, 199, 229-236. DOI: [10.1016/j.scienta.2015.12.047](https://doi.org/10.1016/j.scienta.2015.12.047). Publisher article. [V-P]
12. Magwaza, L. S., & Tesfay, S. Z. (2015). A Review of Destructive and Non-destructive Methods for Determining Avocado Fruit Maturity. *Food and Bioprocess Technology*, 8(10), 1995-2011. DOI: [10.1007/s11947-015-1568-y](https://doi.org/10.1007/s11947-015-1568-y). Publisher article. [V-P]
13. Estrada, J. S., Demarco, R., Johnson, C. M., Zañartu, M., Fuentes, A., & Auat Cheein, F. (2025). A multi-spectral and hyperspectral image dataset for evaluating chemical traits and the water status of avocado, olive and grape through leaf dehydration under laboratory conditions. *Scientific Reports*, 15, 2973. DOI: [10.1038/s41598-025-85714-8](https://doi.org/10.1038/s41598-025-85714-8). Publisher article. [V-P]

B.4 Pigment and peel-colour biology

14. Cox, K. A., McGhie, T. K., White, A., & Woolf, A. B. (2004). Skin colour and pigment changes during ripening of 'Hass' avocado fruit. *Postharvest Biology and Technology*, 31(3), 287-294. DOI: [10.1016/j.postharvbio.2003.09.008](https://doi.org/10.1016/j.postharvbio.2003.09.008). Publisher article. [V-P]
15. Sibeko, D. S., Shikwambana, K., & Mathaba, N. (2024). Effect of ripening temperature on early-season 'Hass' avocado fruit exocarp colour development and pigmentation during ripening. *Horticultural Science*, 51(2), 152-159. DOI: [10.17221/72/2023-HORTSCI](https://doi.org/10.17221/72/2023-HORTSCI). Publisher article. [V-P]
16. Ashton, O. B. O., Wong, M., McGhie, T. K., Vather, R., Wang, Y., Requejo-Jackman, C., Ramankutty, P., & Woolf, A. B. (2006). Pigments in Avocado Tissue and Oil. *Journal of Agricultural and Food Chemistry*, 54(26), 10151-10158. DOI: [10.1021/jf061809j](https://doi.org/10.1021/jf061809j). Publisher article. PubMed record. [V-P]

B.5 Multimodal ripening and dynamic models

17. Gwanpua, S. G., Qian, Z., & East, A. R. (2018). Modelling ethylene regulated changes in 'Hass' avocado quality. *Postharvest Biology and Technology*, 136, 12-22. DOI: [10.1016/j.postharvbio.2017.10.002](https://doi.org/10.1016/j.postharvbio.2017.10.002). Publisher article. [V-P]
18. Sierra, N. M., Londoño, A., Gómez, J. M., Herrera, A. O., & Castellanos, D. A. (2019). Evaluation and modeling of changes in shelf life, firmness and color of 'Hass' avocado depending on storage temperature. *Food Science and Technology International*, 25(5), 370-384. DOI: [10.1177/1082013219826825](https://doi.org/10.1177/1082013219826825). Publisher article. [V-P]
19. van der Sman, R. G. M., Guo, X., & Pedreschi, R. (2026). Physics-Informed Neural Network for prediction of post-harvest firmness of avocados. *Postharvest Biology and Technology*, 231, 113898. DOI: [10.1016/j.postharvbio.2025.113898](https://doi.org/10.1016/j.postharvbio.2025.113898). Publisher article. [V-P]
20. Guo, X., Harbers, C. J., Heuer, H. E. J. M., Min, Q., Liu, Y., Woltering, E. J., Marcelis, L. F. M., & van der Sman, R. G. M. (2026). A generic scientific machine learning framework for fruit and vegetable quality prediction. *Postharvest Biology and Technology*, 234, 114103. DOI: [10.1016/j.postharvbio.2025.114103](https://doi.org/10.1016/j.postharvbio.2025.114103). Publisher article. [V-P]
21. Mishra, P., Paillart, M., Meesters, L., Woltering, E., Chauhan, A., & Polder, G. (2021). Assessing avocado firmness at different dehydration levels in a multi-sensor framework. *Infrared Physics & Technology*, 118, 103901. DOI: [10.1016/j.infrared.2021.103901](https://doi.org/10.1016/j.infrared.2021.103901). Publisher article. [V-P]
22. Melado-Herreros, A., Nieto-Ortega, S., Olabarrieta, I., Gutiérrez, M., Villar, A., Zufía, J., Gorretta, N., & Roger, J.-M. (2021). Postharvest ripeness assessment of 'Hass' avocado based on development of a new ripening index and Vis-NIR spectroscopy. *Postharvest Biology and Technology*, 181, 111683. DOI: [10.1016/j.postharvbio.2021.111683](https://doi.org/10.1016/j.postharvbio.2021.111683). Publisher article. [V-P]

23. Hernández, I., Uarrota, V., Paredes, D., Fuentealba, C., Defilippi, B. G., Campos-Vargas, R., Meneses, C., Hertog, M., & Pedreschi, R. (2021). Can metabolites at harvest be used as physiological markers for modelling the softening behaviour of Chilean 'Hass' avocados destined to local and distant markets? *Postharvest Biology and Technology*, 174, 111457. DOI: [10.1016/j.postharvbio.2020.111457](https://doi.org/10.1016/j.postharvbio.2020.111457). [Institutional publication record](#). [V-P]

B.6 Dataset and code records

24. Xavier, P., Rodrigues, P., & Silva, C. L. M. (2024). 'Hass' Avocado Ripening Photographic Dataset (Version 1) [Data set]. *Mendeley Data*. DOI: [10.17632/3xd9n945v8.1](https://doi.org/10.17632/3xd9n945v8.1). [Dataset landing page](#). Licence: CC BY 4.0. [V-P, R]
25. Waseem, M. S., Sharma, N., & Hefeeda, M. (n.d.). *RipeTrack: Assessing Fruit Ripeness using Smartphones* [Code repository and external dataset links]. GitHub. Retrieved 30 July 2026 from <https://github.com/ShahzaibWaseem/RipeTrack>. No explicit code or dataset licence was located during this audit. [R, U]
26. Varga, L. A., Makowski, J., & Zell, A. (n.d.). *deephs_fruit: Measuring the ripeness of fruit with Hyperspectral Imaging and Deep Learning* [Code and data repository]. GitHub. Retrieved 30 July 2026 from https://github.com/cogsys-tuebingen/deephs_fruit. [Dataset file server](#). No explicit repository or dataset licence was located during this audit. [R, U]
27. Estrada, J. S. (2024). A multi-spectral and hyperspectral image dataset for evaluating the health status of avocado, olive and vineyard (Version 2) [Data set]. *figshare*. DOI: [10.6084/m9.figshare.26950660.v2](https://doi.org/10.6084/m9.figshare.26950660.v2). [Dataset landing page](#). The item-level dataset licence was not retrievable during this audit and must be confirmed on the repository record before reuse. [R, U]

B.7 Citation and access gaps that remain open

Item	Verified	Unresolved action
RipeTrack paper	Title, authors, journal, volume, issue, pages, year, DOI	Obtain explicit terms for code, pretrained weights, and each externally hosted dataset before redistribution
DeepHS Fruit paper and server	Paper citation, repository, modalities, public file server, advertised file sizes	Obtain an explicit data/code licence; audit annotation-to-fruit identifiers before modeling
Estrada leaf dataset	Article citation, repository DOI, modalities, counts	Confirm the separate figshare item licence; article CC BY-NC-ND does not substitute for dataset terms
Han et al. HSI data	Paper citation and "reasonable request" availability	Request data and written reuse terms from the corresponding author
Lee et al. smartphone data	Paper citation and "reasonable request" availability	Request images, fruit IDs, batch IDs, and explicit reuse terms
Davur et al. HSI data	Paper citation and reported IP restriction	Resolve intellectual-property permission; no public raw-data release was identified
Pinto et al. HSI data	Paper citation and seven-fruit design	Contact authors if raw cubes are required; no downloadable dataset was identified
Bratu et al. apple data	Paper citation and article licence	Contact authors for raw gas time series and multispectral images; no machine-readable release was identified

Appendix C - Public and Potentially Obtainable Datasets

Dataset-audit date: 30 July 2026

Evidence and status labels

- **[V-P] Primary-source verified:** checked against the dataset repository, publisher, or associated paper.
- **[V-L] Local-full-text verified:** checked in a supplied local PDF.
- **[A] Audit assessment:** quality, leakage, or recommended-use assessment derived from the published protocol.
- **[U] Unresolved:** licence, access, grouping key, or raw-data status needs confirmation.

Access-status vocabulary

- **Open and licensed:** downloadable and accompanied by an explicit reusable dataset licence.
- **Publicly linked; licence unresolved:** files are linked publicly, but no explicit dataset reuse licence was found.
- **Request-only:** data are available only by contacting the authors.
- **Unavailable:** the article states the data are unavailable or no raw release could be found.
- **Domain-adjacent:** public data exist but do not measure the target object or target outcome.

“Publicly visible” and “licensed for reuse” are not synonyms. Article licences and repository/code licences must not be assumed to cover separate raw datasets.

C.1 Dataset decision table

Dataset	Target and scale	Modalities and labels	Licence and access	Quality assessment	Recommended use	Evidence
‘Hass’ Avocado Ripening Photographic Dataset , DOI 10.1763/2/3xd9n945v8.1	478 independent Hass fruits; 14,710 labeled 800×800 JPEGs; two opposite sides photographed daily	RGB photographs; fruit/sample number; storage group; day; side; five-stage ripening index; timestamps supporting days-left derivation	Open and licensed: CC BY 4.0	High for controlled RGB longitudinal baselines. Strong sample count and explicit fruit IDs; the associated paper states that each fruit/sample was assigned wholly to one model partition. Labels are researcher-assigned visual/tactile stages, not destructive chemistry for every fruit: only eight separate fruits were used for the initial dry-matter assessment. DSLR lightbox imagery creates a domain gap to uncontrolled smartphones.	Acquire first. Split by sample number. Keep all days and sides from one fruit in one fold. Use storage group for domain-shift evaluation; do not randomly split images.	[V-P, A]
RipeTrack datasets	48 total fruits across seven fruit/cultivar groups; avocado subset is 10 organic + 3 Hass; 1,913 HSI and 1,144 ethylene readings overall; 3,865 mobile RGB-NIR pairs	400-1000 nm HSI, RGB/NIR phone captures, ethylene, time/ripeness and remaining-life labels	Publicly linked; licence unresolved. GitHub links to Google Drive; no explicit dataset licence found	Scientifically relevant but biologically small for avocado. Effective avocado n=13, Hass n=3; repeated images must not be treated as independent fruit. Reconstruction split grouping is unclear.	Use only after written licence clarification. If obtained, preserve fruit identity, day, cultivar, phone, illumination, and capture distance. Suitable for transfer-learning and sensor-comparison experiments, not as a definitive external test set.	[V-P, V-L, A, U]
DeepHS Fruit v2 and file server	Fruits include avocado, kiwi, persimmon/kaki, papaya, and mango across four measurement series; avocado archive advertised as 72 GB	HSI from Specim FX10, INNO-SPEC Redeye 1.7, and Corning microHSI 410; destructive flesh firmness, sugar content, and overall ripeness labels	Publicly linked; licence unresolved. Direct/torrent downloads exist; no explicit dataset or repository licence found	Potentially high scientific value, high operational cost. Multiple cameras and destructive labels enable spectral-domain work. v1 had 1,038 avocado recordings but only 180 destructively labeled avocado recordings. Front/back/repeated records create leakage risk unless grouped by fruit.	Do not download for routine use until licence and storage budget are approved. First inspect the small annotation archive and verify stable fruit IDs, camera IDs, series IDs, sides, and label completeness. Use grouped fruit/series/camera validation.	[V-P, A, U]
Estrada et al. avocado/olive/grape leaf dataset , DOI 10.6084/m9.figshare.26950660.v2	106 avocado leaves, 111 olive leaves, 106 grape leaves; five dehydration stages	Five-band multispectral TIFFs; 350-2500 nm spectra; weight; chlorophyll/SPAD; nitrogen; dry- and fresh-basis fuel moisture	Domain-adjacent; item licence unresolved. Article links the repository, but this audit could not verify the separate figshare item licence	High for leaf water spectroscopy; low for fruit ripeness. Leaf anatomy and forced oven dehydration differ materially from avocado peel/mesocarp ripening. Paper reports 2,600 avocado TIFFs although 106 leaves × 5 stages × 5 bands implies 2,650, requiring a file-manifest audit.	Use only for wavelength-selection, spectral-index code tests, or water-sensitivity method development. Never use as a fruit-ripeness train/test set. Confirm item licence and missing-file pattern first.	[V-P, A, U]

Dataset	Target and scale	Modalities and labels	Licence and access	Quality assessment	Recommended use	Evidence
Han et al. 2023 Hass/Shepard HSI	316 Hass and 160 Shepard fruits; one HSI per fruit after cold storage	462 bands, 388-1005 nm; days to full ripeness; cultivar; firmness threshold used to define ripeness	Request-only. Publisher states data are available from the corresponding author on reasonable request	Strong sample count. Fruit-level PLSR calibration/test design is useful. The DCNN used randomly split subimages, allowing likely same-fruit leakage between partitions.	Request for a confirmatory HSI benchmark. Prefer the one-record-per-fruit ROI spectra or split all subimages by fruit. Retain cultivar and harvest-date groups.	[V-P, A]
Davur et al. 2023 Hass longitudinal HSI	80 independent Hass fruits; 551 daily HSI; paper reports 44,096 subimages	Pika XC2/Resonon HSI, 462 bands, 388.9-1005.3 nm; days until ripeness	Unavailable due to intellectual-property arrangements	Valuable longitudinal design but unsuitable published split. Patches from each original HSI were allocated across training, validation, and test, causing direct same-image and same-fruit leakage. The reported patch totals and described allocation also require arithmetic reconciliation.	Use as a protocol and leakage caution, not as a performance benchmark. Seek IP permission only if raw fruit/day IDs and a new grouped analysis are possible.	[V-P, V-L, A]
Lee, Li, and Ma 2025 smartphone dataset	140 Hass fruits in two 70-fruit batches; 10 views per fruit; 1,400 images; eight-day study	Smartphone RGB; destructive firmness; fresh/rotten internal labels; storage day and batch	Request-only. Data-availability statement says available on reasonable request	Useful multimodal ground truth, but main random image split likely leaks fruit identity. Cross-batch results are more credible and show substantial direction-dependent domain shift.	Request images with fruit and batch identifiers. Rebuild grouped fruit splits and leave-one-batch-out tests. Use for smartphone-domain validation, not as a direct comparison to image-random headline metrics.	[V-P, A]
Pinto, Rueda-Chacón, and Arguello 2019 Hass HSI	Seven Hass fruits followed for 10 days; three maturity categories	HSI; PCA; NDVI, RVI, PRI, CIE L*a*b*, and TGI analyses	Unavailable. Article is accessible, but no raw dataset download was identified	Low statistical power. Useful for feature ideas only; seven fruits cannot support a generalizable model.	Cite for exploratory spectral indices and Colombian context. Do not use its results as an expected-accuracy benchmark. Contact authors only if raw cubes would aid method debugging.	[V-P, A]
Bratu et al. 2021 apple gas and imaging study	20 apples, five per cultivar, six weekly time points over 35 days	LPAS ethylene and ethanol; eight-filter visible multispectral imaging; external appearance	Unavailable. No public machine-readable raw release identified	Domain mismatch and small n. Gas system is precise, but the imaging analysis was mostly inconclusive. Apple gas trajectories are not avocado kinetic parameters.	Use to design gas-chamber controls and synchronized measurements. Do not train an avocado model or infer avocado rate constants from it.	[V-P, V-L, A, U]

C.2 Priority acquisition plan

Priority 1 - acquire and audit now

‘Hass’ Avocado Ripening Photographic Dataset (Mendeley Data).

Reasons:

- explicit CC BY 4.0 dataset licence;
- 478 biological fruits;
- stable sample number suitable for grouped validation;
- longitudinal days and opposite-side views;
- controlled storage and imaging metadata; and
- manageable RGB format compared with tens of gigabytes of HSI.

Minimum intake checks:

1. verify archive hashes and preserve the untouched download;
2. retain the dataset DOI, version, licence text, and retrieval date;
3. reconcile 14,710 image files with the spreadsheet;
4. check for missing/duplicate filenames and sample numbers;
5. quantify samples per storage group, day, stage, and side;
6. confirm no sample number occurs in more than one partition;
7. document whether shelf-life endpoint derivation uses stage 5 while the narrative also describes stage 4 as the end of optimal shelf life; and
8. distinguish controlled DSLR performance from smartphone performance.

Priority 2 - permission and metadata first

RipeTrack and **DeepHS Fruit** should not enter the project's reusable data corpus until the rights holder supplies or points to explicit licence terms. Before downloading large files:

- request data and code reuse/redistribution terms;
- request a data dictionary and stable fruit identifiers;
- inspect small annotations/manifests first;
- estimate storage and preprocessing costs;
- confirm whether pre-trained weights have separate terms; and
- record any restrictions on commercial use, redistribution, or derived models.

Priority 3 - author requests

Request the Han et al. and Lee et al. datasets because their independent-fruit counts and ground truth could materially improve external validation. A request should ask for:

- raw files and checksums;
- fruit, batch, cultivar, orchard/harvest, date, side/view, and instrument identifiers;
- exact label derivation and measurement units;
- calibration/white-reference files for HSI;
- permission to analyze, publish derived statistics, train models, and redistribute only what the licence permits; and
- citation and acknowledgement requirements.

Priority 4 - methods-only sources

Use Davur et al., Pinto et al., and Bratu et al. as design and failure-mode references unless rights and raw data become available. Their current published validation designs or species/sample limitations prevent their headline performance from serving as a reliable target benchmark.

C.3 Dataset quality rubric

Each candidate dataset should be scored against the following criteria before model development:

Criterion	Acceptable evidence
Biological independence	Count of distinct fruits, trees, batches, orchards, seasons, and cultivars-not merely images or patches
Grouping keys	Stable fruit/sample ID and identifiers for repeated day, side, patch, instrument, and batch
Ground-truth traceability	Measurement protocol, unit, instrument, calibration, operator, and timing relative to imaging

Criterion	Acceptable evidence
Label validity	Clear distinction among firmness, eating ripeness, internal rot, days to threshold, dry matter, moisture, and subjective colour stage
Leakage control	All records derived from one fruit confined to one fold; no patch-, side-, or day-level random split across folds
External validity	Independent harvest batch, orchard, season, storage condition, phone/camera, or geographic source
Licensing	Explicit dataset licence and attribution terms; no inference from “public repository” or article licence
Completeness	File manifest reconciles with metadata; missing/corrupt records and exclusions documented
Calibration data	White/dark references for spectral instruments; colour references and fixed exposure for RGB where available
Reproducibility	Versioned annotations, checksums, documented preprocessing, and reproducible train/test assignment

C.4 Required split policy for every adopted dataset

Let g_i denote the biological fruit identifier for observation i . Every image, spectral patch, side, wavelength file, phone capture, and day derived from fruit g_i must belong to exactly one of training, validation, or test:

$$\begin{aligned} \mathcal{G}_{\text{train}} \cap \mathcal{G}_{\text{val}} &= \mathcal{G}_{\text{train}} \cap \mathcal{G}_{\text{test}} = \\ \mathcal{G}_{\text{val}} \cap \mathcal{G}_{\text{test}} &= \text{varnothing}. \end{aligned}$$

Where sufficient metadata exist, the project should add a higher-level external test split by harvest batch, orchard, season, storage treatment, or device. Image-level random splitting is prohibited for longitudinal fruit datasets because it estimates recognition of previously seen fruit and acquisition conditions rather than generalization to a new fruit. **[A]**

C.5 Licensing and availability caveats

- Mendeley Hass RGB data:** CC BY 4.0 is verified at the dataset record. Attribution and version/DOI retention are required.
- RipeTrack:** public repository and Google Drive links are verified; explicit code/data licence is not. Do not redistribute or publish derived subsets until permission is documented.
- DeepHS Fruit:** public direct and torrent files are verified; explicit code/data licence is not. Its 72 GB avocado archive should not be acquired merely because it is reachable.
- Estrada figshare data:** the associated Scientific Reports article is CC BY-NC-ND 4.0, but the dataset is a separate repository object. The separate item licence must be checked; do not substitute the article licence.
- Request-only studies:** “available on reasonable request” is an access route, not a licence. Written terms should accompany the transfer.
- Unavailable/IP-restricted studies:** do not attempt to reconstruct or redistribute raw data from figures or supplementary screenshots. Use only reported aggregate findings within normal citation limits.
- Article images and equipment photographs:** article open-access terms may differ from dataset terms and third-party photo credits. Each proposed reused image needs its own rights check.

C.6 Dataset-to-research-task mapping

Research task	Best current source	Why	Required safeguard
RGB ripeness-stage baseline	Mendeley Hass photographic dataset	Largest verified, licensed longitudinal fruit dataset in this audit	Fruit-grouped split; report controlled-lightbox domain
RGB remaining-life baseline	Mendeley Hass photographic dataset	Day and endpoint metadata allow days-left derivation	Freeze endpoint definition before modeling; avoid post-outcome leakage
Smartphone external validation	Lee et al., if obtained	Smartphone captures plus firmness/internal quality	Request fruit IDs; leave-one-batch-out evaluation
VIS/NIR days-to-ripen regression	Han et al., if obtained	476 independent fruits and one HSI per fruit	Whole-fruit split; separate cultivars and harvest dates
HSI representation learning	DeepHS Fruit, if licensed	Multiple cameras, fruit types, and destructive labels	Fruit/series/camera grouped folds; annotation audit
RGB-NIR phone reconstruction	RipeTrack, if licensed	Paired mobile data and reconstruction pipeline	Confirm fruit grouping and device/illumination metadata
Water-sensitive wavelength prototyping	Estrada leaf dataset, if licensed	Dense 350-2500 nm spectra with measured dehydration	Treat as leaf-only auxiliary evidence; validate on fruit tissue
Gas-chamber protocol	Bratu et al. paper	Detailed LPAS chamber and temporal gas workflow	New avocado calibration; selective sensors and environmental controls

Appendix D - Prototype Hardware Catalogue

Project: Multimodal non-destructive Hass avocado ripeness characterisation **Procurement evidence date:** 2026-07-30
Status: proposal summary; the audited procurement records remain authoritative

D.1 Purpose and controlling records

This appendix condenses the proposed hardware into an implementation-oriented catalogue. It should be read with the following local project records:

- Audited bill of materials
- Detailed equipment catalogue
- Procurement and specification source ledger
- Prototype architecture
- Local product-image licence and attribution record

The bill of materials is the controlling item-level record for manufacturer, exact model or manufacturer part number, quantity, price, currency, price status, verification date, supplier, source URL, specification, intended use, and limitation. A blank price means that no current price was verified; it does not mean zero cost.

D.2 Procurement tiers and budgets

Configuration	Scope	Budget basis	Total
MVP priced core	Live-priced or current-MSRP controller, imaging, reference target, environmental sensors, CO2, MOX gas trend, mass, ADC, storage, power, chamber, and silicon detector	Audited BOM rows in MVP_priced_core	US\$443.67
MVP planning configuration	MVP priced core plus explicitly labelled fabrication, optical, fan, tubing, storage-media, calibration-weight, and mounting allowances	Audited BOM rows in MVP_priced_core and MVP_allowances	US\$723.62
Research-grade using shared core instruments	MVP plus discrete SWIR feasibility channels, analytical-chamber/plumbing allowances, source/probe/reference-standard allowances, refractometer, penetrometer, and destructive-assay consumables	Assumes spectrometer, reference ethylene analyser, moisture analyser/oven, analytical balance, and chemistry facilities are university-owned	US\$6,591.51
Research-grade buy-new baseline proxy	Shared-instrument configuration plus F-900, OHAUS MB120, and a historical NIRQuest+1.7 package proxy	Budgetary comparison only	US\$43,466.01
Mid-tier compact spectrometer option	Hamamatsu C11708MA alone	Observed authorised-distributor price	€650.53 ex VAT

These totals exclude tax, UAE shipping, duty, institutional procurement fees, calibration certificates and gases, optional rows, and quote-only rows unless expressly stated. The **US\$43,466.01 buy-new value is not a vendor quotation**. Its NIRQuest component is transparently based on a 2023 federal contract record because the manufacturer currently quotes by optical configuration. A current written quotation is required before funding or purchase decisions.

If oil-sensitive bands near approximately 1720 and 2300 nm are a primary scientific objective, a 900-2500 nm system must be quoted. The baseline NIRQuest+1.7 option ends near 1650 nm and therefore does not cover those regions.

D.3 Minimum viable prototype

D.3.1 Control, imaging, and storage

Function	Selected model / MPN	Purpose	Principal limitation
Sensor controller	Espressif ESP32-S3-DevKitC-1-N8R8	LED sequencing, environmental and mass acquisition, timestamps, microSD records	Integrated ADC is not the precision optical acquisition path
Image computer	Raspberry Pi Zero 2 W, ordering code SC1176	Camera control and image metadata	Reseller availability may exceed official MSRP
RGB camera	Raspberry Pi Camera Module 3 standard; Adafruit PID 5657	Repeatable peel images	Camera RGB is device- and illumination-dependent; all automatic settings must be locked
Visible reference	Calibrite ColorChecker Classic Mini CCC-MINI	Colour calibration and drift checks	Not an NIR diffuse-reflectance standard
Local logger	Adafruit microSD breakout PID 254 plus qualified endurance card	Append-only raw records	Card authenticity, endurance, and corruption handling require verification

D.3.2 Environment, respiration, and broad gas response

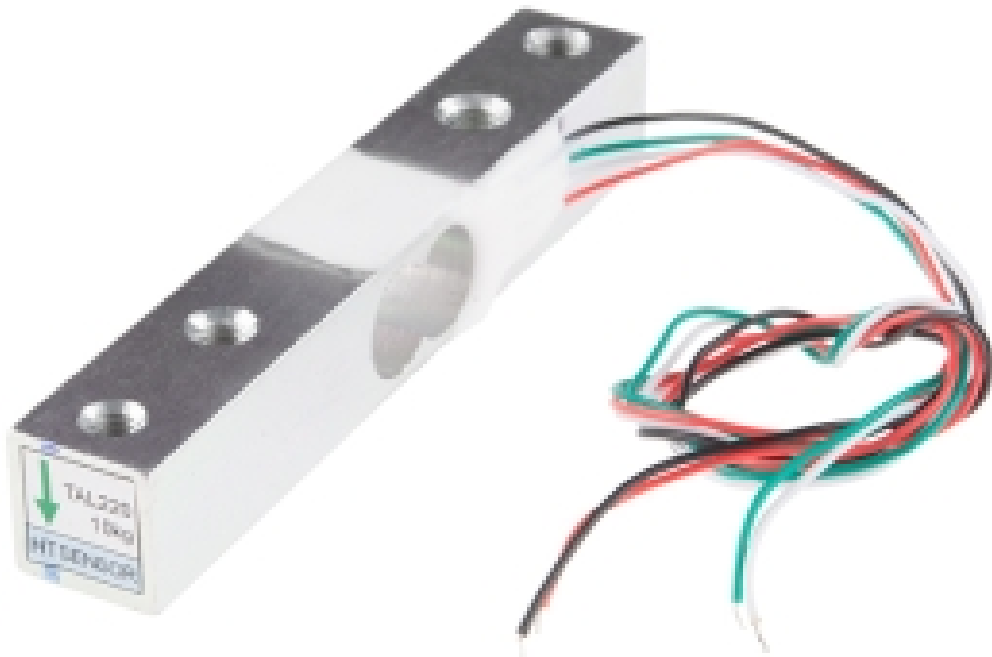
Function	Selected model / MPN	Defensible measurement	Claim that is not justified
Temperature /RH	Sensirion SHT45-AD1B-R2; Adafruit PID 5665	Chamber-air temperature and relative humidity	Internal fruit water concentration
CO ₂	Sensirion SCD30; Adafruit PID 4867	NDIR CO ₂ concentration and a corrected sealed-interval accumulation slope	Ethylene, or respiration rate without chamber-volume, leak, pressure, temperature, and response corrections
Broad gas feature	Bosch BME688; Adafruit PID 5046	Conditioned metal-oxide gas-resistance trajectory	Ethanol, ethylene, or any other individual VOC concentration

The SCD30 is specified for 400-10,000 ppm. The accumulation interval must be shortened or the chamber enlarged/flushed before the signal exceeds its specified range. The BME688 is a humidity-, temperature-, baseline-, heater-history-, and mixture-dependent sensor. It is an optional multivariate odour feature, not a chemical analyser.

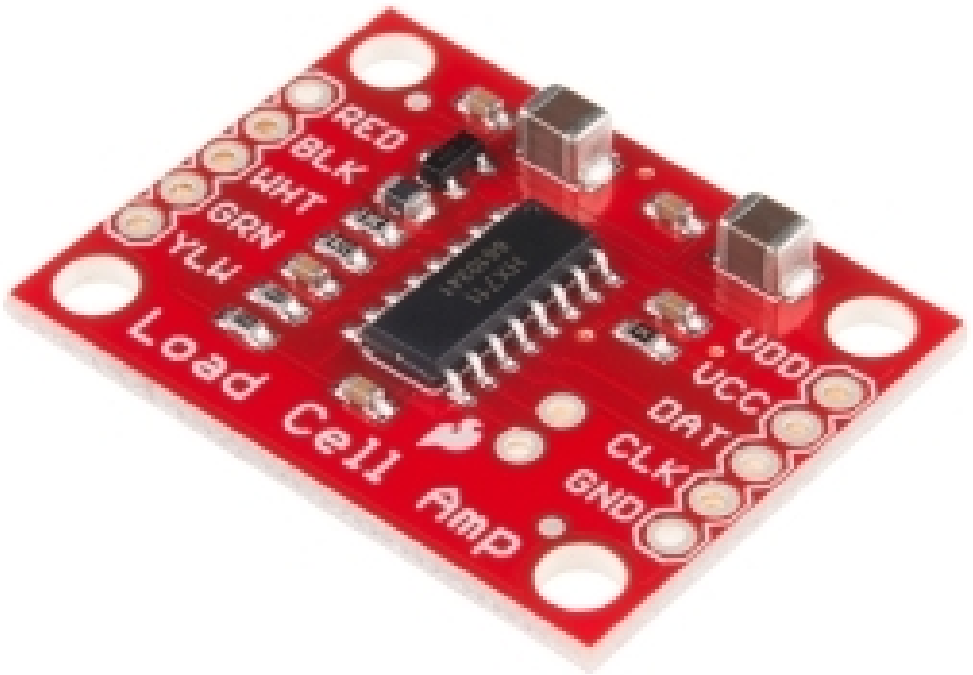
D.3.3 Mass

Function	Selected model / MPN	Purpose	Principal limitation
Force transducer	SparkFun TAL220 10 kg straight-bar load cell, SEN-13329	Fruit mass trajectory	Creep, temperature, mounting, off-axis loading, and cradle contact affect accuracy
Bridge acquisition	SparkFun HX711 board, SEN-13879	Load-cell excitation/readout	A 24-bit output word does not imply 24-bit system accuracy

The assembled mechanism must be calibrated with traceable or documented check masses. Tare, stable-window criteria, temperature, fan/pump state, and before/after check weights should be recorded for each session.



SparkFun TAL220 10 kg load cell



SparkFun HX711 load-cell amplifier

The images above are exact-model SparkFun product photographs. Attribution, licence, original URLs, and SHA-256 hashes are recorded in the local image register.

D.3.4 MVP optical head

Channel	Exact proposed part	Intended feature	Limitation
Detector	Vishay BPW34	Sequential reflected-light detection from 660 to 970 nm	Silicon response ends near 1100 nm
660 nm	Roithner LED660N-03	Chlorophyll-sensitive visible reflectance	Also affected by scattering, skin, geometry, and temperature
740 nm	Roithner LED740-03AU	Red-edge/pigment-scattering transition	Not a unique molecular concentration
850 nm	Roithner LED850-03UP	Tissue/skin scattering reference	Not a water or oil concentration
970 nm	Roithner LED970D-03	Weak water-sensitive O-H feature	Must not be reported as laboratory moisture percentage without calibration
ADC	Adafruit ADS1115 PID 1085	Slow transimpedance-amplifier channels	Does not provide a microspectrometer line-array readout

The four-LED head is intended to generate controlled predictive features. It is not a spectrometer and cannot observe stronger avocado-relevant features near 1200, 1450, 1700, or 2300 nm. It requires:

- sequential constant-current LED drive;
- a low-input-bias transimpedance amplifier;

- fixed source-fruit-detector geometry and baffling;
- dark measurements before every sequence;
- a stable diffuse reference at the beginning and end of every session;
- recorded LED current, gain, integration time, sensor temperature, saturation, and component bin.

The exact LED MPNs were verified in the manufacturer's current part/price list, but their combined US\$50 value is a planning allowance pending a written bin-specific quotation.

D.3.5 Chamber, mixing, and power

Function	Selected model / MPN	Purpose	Principal limitation
MVP chamber shell	Integra H12106HCF-6P-P10	Repeatable one-fruit enclosure and sensor mounting	Polycarbonate, gasket, adhesives, coatings, and modified ports can outgas or adsorb volatiles
Mixing fan	Noctua NF-A4x10 5V	Fixed-duty headspace mixing	Heat and airflow alter gas and moisture transport; switch off during weighing
Low-voltage supply	Mean Well GST60A12-P1J	External 12 V, 5 A supply	IEC input lead and fused/switchable downstream distribution are required
5 V rail	SparkFun AP63357 breakout COM-21255	Regulated digital/fan rail	Observe current and thermal derating

The enclosure's original environmental rating applies before drilling or modifying it. An empty-chamber blank, leak test, dose/recovery test, and cleaning/cure protocol are mandatory. A borosilicate or stainless vessel with PTFE-faced seals is preferred for quantitative volatile work.

D.4 Optional and bridge equipment

D.4.1 AS7341 colour-spectral breakout

The ams OSRAM AS7341 breakout, Adafruit PID 4698, offers eight visible bands plus broad NIR, clear, and flicker channels. It can enrich surface-pigment features but is neither a continuous spectrum nor a water/oil NIR analyser. It was out of stock at the audited US\$18.95 supplier price and should not block the MVP.

D.4.2 Low-cost electrochemical ethylene

SPEC Sensors 970-650 (DGS2-C2H4) is an optional **bridge sensor**, not the project's ethylene ground truth. The manufacturer's cross-sensitivity information shows responses to ethanol, hydrogen sulfide, sulfur dioxide, nitric oxide, and formaldehyde. Avocado headspace is a mixed, humid matrix, so the module may be used only after comparison against a reference analyser across:

- ethylene calibration level;
- temperature and RH;
- CO₂ and ethanol;
- fruit load and ripeness stage;
- response/recovery time;
- baseline drift and repeated cycles.

Report bias, precision, selectivity, limit of detection/quantification, and uncertainty before interpreting its output as concentration.

D.4.3 Low-cost pump

The SparkFun 12 V vacuum pump may flush a prototype loop. Its wetted materials are not specified for quantitative volatile recovery, so it must remain outside the analytical reference path unless blank, adsorption, and recovery performance are established.

D.5 Research-grade optical configuration

D.5.1 Compact 640-1050 nm option

The Hamamatsu C11708MA is a 256-pixel, 640-1050 nm miniature spectrometer with typical 15 nm spectral resolution. It is a bare analogue head requiring optics, timing, buffering, ADC acquisition, shielding, and dark/white calibration. It can improve pigment, scattering, and weak 970 nm measurements but cannot observe the stronger SWIR water/lipid regions.

The smaller C14384MA-01 is a quote-only alternative with higher sensitivity near 1000 nm. No current authorised price was verified.

D.5.2 Calibrated SWIR option

The proposed research baseline is a cooled InGaAs instrument such as Ocean Insight NIRQUEST+1.7-XX, configured by quotation with:

- a stabilised tungsten-halogen source;
- matched fibres;
- repeatable interactance and reflectance probe fixtures;
- slit/grating selected for the required resolution and throughput;
- certified diffuse reflectance standard;
- wavelength and photometric checks.

Its 900-1650 nm range captures useful 1200 and 1450 nm information. An extended-range NIRQuest+2.5 or equivalent should be quoted for 1720/2300 nm oil-sensitive work.

Three Marktech MTPD1346D-100 InGaAs photodiodes and Roithner LED1200S-03/LED1450S-03 emitters are budgeted as a discrete-band feasibility step. These channels require low-noise analogue electronics and dark/temperature correction and do not replace spectroscopy.

D.6 Research-grade headspace configuration

Function	Proposed equipment	Procurement status
Reference ethylene	Felix Instruments F-900 or validated GC method	Prefer university-shared; F-900 observed at US\$10,495
Chamber	Custom known-volume borosilicate/stainless vessel with PTFE-faced seals	US\$1,000 design allowance; engineering/RFI required
Tubing	Cole-Parmer PTFE UX-50119-48	US\$367.50 for 50 ft at audit
Research pump	KNF NMP 830, configuration with suitable wetted materials	Quote only
Flow control	Low-dead-volume meter, valves, fittings, filters	US\$250 allowance pending frozen flow specification

The F-900 manufacturer’s PPB-channel specification is 0-10 ppm with 0.001 ppm resolution, stated 25 ppb detection limit, and stated accuracy of 5\% \pm 0.025 ppm. The applicable calibration and interference-removal protocol must still be followed. Reference-grade equipment does not eliminate the need for humid-headspace method validation.

D.7 Destructive ground truth

Measurement	Equipment	Procurement decision	Key limitation
Soluble solids	ATAGO PAL-1, Cat. No. 3810	US\$330 direct price	Brix is concentration and includes effects of water loss and non-sugar solutes
Firmness	QA Supplies FT327, SKU 2006060	US\$285	Fresh harvest fruit may exceed range; probe/site/speed/depth protocol determines repeatability
Moisture/dry matter	OHAUS MB120 AM, material 30241165, or validated oven method	Prefer laboratory-owned; 2026 authorised MAP US\$6,400.50	Halogen endpoint must be validated against the adopted reference method
Starch and oil	Validated laboratory assay/extraction facilities	Institutional method and safety plan required	No product number or performance is asserted before method selection

The proposal must retain moisture/dry matter, mass, Brix, starch, firmness, and oil as separate variables. Brix alone cannot distinguish biochemical sugar formation from concentration caused by water loss.

D.8 Procurement sequence

1. Confirm university-owned spectrometer, F-900/GC, moisture analyser/oven, analytical balance, chemistry facilities, and gas-calibration resources.
2. Order the MVP priced-core items.
3. Obtain the exact Roithner LED quotation and select wavelength bins.
4. Freeze chamber dimensions, optical fixture, analogue gains, plumbing sizes, and safety design.
5. Purchase fabrication and consumables against final drawings rather than allowances.
6. Complete incoming inspection, serial/lot records, calibration, blank, leak, recovery, optical repeatability, load-cell creep, and clock-synchronisation tests.
7. Collect biological data only after the acceptance criteria in the prototype architecture are satisfied.

D.9 Documentation and non-quotation notice

Every purchase should archive the supplier quotation, access date, currency, tax/freight, lead time, exact MPN, datasheet revision, received serial/lot, calibration certificate, incoming test, and warranty. The project's source ledger records the evidence used at proposal time.

This appendix is a technical planning document. It is not a request for quotation, offer, warranty of availability, or representation of landed cost. Prices and stock must be reconfirmed by the purchasing institution.

Appendix E - Candidate Mathematical and Scientific-ML Models

E.1 Status and purpose

Every model in this appendix is **PROPOSED** unless a cited source is stated. The equations are candidate representations to compare against data. They are not asserted as complete avocado physiology, and no biological rate constant is estimated from the public RGB dataset because that dataset lacks the required states and interventions.

Prior avocado kinetic modeling already exists. In particular, Gwanpua et al. modeled autocatalytic ethylene and linked ethylene to softening and colour using temperature-dependent kinetics ([DOI 10.1016/j.postharvbio.2017.10.002](https://doi.org/10.1016/j.postharvbio.2017.10.002)). Sierra et al. also reported temperature-dependent quality kinetics ([DOI 10.1177/1082013219826825](https://doi.org/10.1177/1082013219826825)). Van der Sman, Guo, and Pedreschi have since applied a physics-informed neural network to postharvest avocado firmness using literature datasets, enzymatic softening, and metabolic-rate structure ([DOI 10.1016/j.postharvbio.2025.113898](https://doi.org/10.1016/j.postharvbio.2025.113898)). Guo et al. also published a generic SciML framework for fruit and vegetable quality prediction ([DOI 10.1016/j.postharvbio.2025.114103](https://doi.org/10.1016/j.postharvbio.2025.114103)). Hernández et al. used metabolites measured at harvest as physiological markers in a model of batch-specific Hass softening across postharvest logistics ([DOI 10.1016/j.postharvbio.2020.111457](https://doi.org/10.1016/j.postharvbio.2020.111457)). The proposed contribution is therefore not to claim the first avocado equation or PINN. It is to observe multiple coupled states under a leakage-resistant, externally validated design and test whether additional complexity is identifiable and useful.

E.2 Notation

Symbol	Meaning	Typical unit	Measurement/latent status
t	Time after defined study baseline	h or d	Direct
T_K	Fruit/chamber absolute temperature	K	Direct air; fruit-core optional
RH	Fractional ambient relative humidity	0-1	Direct
v	Local airflow or mixing condition	m/s or fixed level	Assigned/logged
M	Whole-fruit mass	g	Direct
M_w	Fruit water mass	g	Latent longitudinal; destructive calibration
W	Local or bulk water concentration	mass fraction	Destructive calibration/latent
F	Standardized firmness	N	Destructive reference/latent sentinel state
E	Tissue or production-related ethylene state	chosen normalized or physical unit	Latent unless reference method
C_E	Chamber ethylene concentration	ppm or umol/mol	Reference measurement
C_C	Chamber CO2 concentration	ppm or mol fraction	Direct NDIR
C_{chl}	Peel chlorophyll state	concentration	Destructive subset/latent
A_{anth}	Peel anthocyanin state	concentration	Destructive subset/latent
S_{starch}	Starch mass/state	g or concentration	Destructive
S_{sol}	Soluble-solid mass/state	g	Destructive/model state
M_L	Lipid mass	g	Destructive calibration
R	Dimensionless latent progression state	0-1	Latent
V_h	Free chamber headspace volume	L or m3	Calibrated geometry
A_f	Effective fruit surface area	m2	Geometry estimate
R_g	Universal gas constant	consistent units	Constant
y_i	Sensor observation for channel i	channel-specific	Direct

Units must be frozen and checked before fitting. A model that mixes days and seconds, percentage and fraction, or ppm and mole fraction is invalid even if its optimizer converges.

E.3 Environmental rate modifier

E.3.1 Arrhenius candidate

$$k_j(T_K) = A_j \exp\left(-\frac{E_{a,j}}{R_g T_K}\right)$$

Interpretation: apparent process rate j increases with temperature under an Arrhenius approximation.

Assumptions: a stable mechanism over the studied temperature range; temperature is measured on an appropriate scale; storage history is included; no unmodeled chilling injury or phase change.

Limitation: a narrow temperature range can make A_j and $E_{a,j}$ strongly correlated. Estimate a reference-temperature rate and apparent activation energy, or use a Q10 form, rather than extrapolating.

E.3.2 Q10 candidate

$$k_j(T) = k_j(T_{\text{ref}}) Q_{10,j}^{(T - T_{\text{ref}})/10}$$

This is often easier to estimate across a narrow practical range but remains an empirical approximation. Compare it with Arrhenius behavior on held-out temperature conditions.

E.4 Water transport and total mass

E.4.1 Lumped water-loss ODE

$$\frac{dM_w}{dt} = -k_w(T, RH, v) A_f \left[a_w(M_w, T) - RH \right]$$

Possible rate modifier:

$$k_w(T, RH, v) = k_{w, \text{ref}} \exp\left[-\frac{E_{a,w}}{R_g} \left(\frac{1}{T_K} - \frac{1}{T_{\text{ref}, K}}\right)\right] g(v).$$

Interpretation: water flux is driven by an effective surface-to-air vapor activity difference and affected by temperature, RH, airflow, area, and skin.

Required observations: repeated mass, continuous T/RH, airflow setting, fruit geometry, and destructive moisture/dry matter at matched times.

Non-claim: scale loss is not automatically water loss. Respiration removes carbon, condensation can add mass, and handling deposits can alter the scale.

E.4.2 Total mass balance

$$\frac{dM}{dt} = \frac{dM_w}{dt} - r_C(T, R, E) + r_{\text{deposit}}.$$

If CO₂ production is measured in mol/time, a stoichiometric carbon-loss term can be explored, but the substrate and respiratory quotient must not be assumed without evidence.

E.4.3 Spatial diffusion PDE

$$\frac{\partial W(\mathbf{x}, t)}{\partial t} = \nabla \cdot [D_W(W, T) \nabla W] - q_W,$$

with convective boundary condition

$$-D_W \nabla W \cdot \mathbf{n} = h_m [W_s - W_{\text{eq}}(T, RH)].$$

Boundary conditions: symmetry or zero flux at the geometric center; a convective skin boundary; initial spatial water field from destructive matched fruit or a justified homogeneous approximation.

Identifiability warning: whole-fruit mass and one NIR surface observation cannot identify a spatial diffusion coefficient and boundary coefficient simultaneously. Use spatial destructive samples, tomography, informative priors, or retain the lumped ODE.

E.5 Firmness and softening

E.5.1 First-order model

$$\frac{dF}{dt} = -k_F(T)(F - F_{\infty}).$$

Solution:

$$F(t) = F_{\infty} + (F_0 - F_{\infty})e^{-k_F t}.$$

This is parsimonious and should be a baseline even if it underfits the climacteric transition.

E.5.2 Weibull/stretched-exponential model

$$F(t) = F_{\infty} + (F_0 - F_{\infty}) \exp[-(k_F t)^n].$$

The shape parameter n can express acceleration or deceleration, but it may be confounded with k_F for short trajectories.

E.5.3 Ethylene-modulated softening

$$\frac{dF}{dt} = -k_F(T) \frac{E^h}{K_E + E^h} (F - F_{\infty}).$$

Alternative Michaelis-Menten form:

$$\frac{dF}{dt} = -k_F(T) \frac{E}{K_E + E} (F - F_{\infty}).$$

Required observations: independently calibrated ethylene trajectory and a standardized firmness method. A low-cost cross-sensitive sensor cannot identify this coupling without reference comparison.

E.6 Ethylene production and headspace observation

E.6.1 Autocatalytic tissue/production state

$$\frac{dE}{dt} = k_{E,0}(T) + k_{E,a}(T) \frac{E^h}{K_E + E^h} - k_{E,l}(T)E.$$

This candidate represents basal production, autocatalytic induction, and loss. It is related in concept to published avocado ethylene kinetics but is not a transcription of one fitted source model.

E.6.2 Chamber mass balance

$$\frac{dC_E}{dt} = \frac{m_f}{V_h} r_E(E, T) - k_{\text{leak},E}(C_E - C_{E,\text{amb}}) - k_{\text{ads},E}C_E - q_{\text{sensor},E}.$$

Initial condition: measured ambient/baseline concentration after the defined flush/equilibration protocol.

Boundary/operating conditions: known free headspace, fixed mixing, recorded pressure/temperature, defined seal/vent times, and no instrument saturation.

Measurement model:

$$y_E(t) = b_E + s_E C_E(t - \tau_E) + \epsilon_E(t),$$

where b_E , s_E , and τ_E capture offset, sensitivity, and response delay. Cross-sensitivity terms must be added for the low-cost channel.

E.7 Respiration and CO₂

For a short, well-mixed accumulation interval:

$$r_{\text{CO}_2} \approx \frac{V_h}{m_f} \frac{P}{R_g T_K} \frac{dx_{\text{CO}_2}}{dt}.$$

A fuller balance is:

$$\frac{dC_C}{dt} = \frac{m_f}{V_h} r_C(R, T, E) - k_{\text{leak}, C} (C_C - C_{\text{amb}}).$$

Candidate climacteric response:

$$r_C(R, T, E) = r_{C, \theta}(T) + r_{C, \text{peak}}(T) \exp\left[-\frac{(R - R_{\text{peak}})^2}{2\sigma_R^2}\right].$$

Required corrections: chamber free volume after fruit displacement, temperature, pressure, empty-chamber drift, leakage, sensor response, and range. If the NDIR channel exceeds specification, shorten the interval or change the chamber; do not extrapolate the sensor.

E.8 Peel chlorophyll, anthocyanin, and colour

E.8.1 Chlorophyll loss

$$\frac{dC_{\text{chl}}}{dt} = -k_{\text{chl}}(T) C_{\text{chl}}.$$

E.8.2 Anthocyanin accumulation

$$\frac{dA_{\text{anth}}}{dt} = k_A(T) f_A(R) (A_{\text{max}} - A_{\text{anth}}) - k_{A, d} A_{\text{anth}}.$$

One activation choice is

$$f_A(R) = \frac{R^h}{K_A^h + R^h}.$$

E.8.3 Colour observation

$$\mathbf{y}_{\text{colour}} = g(C_{\text{chl}}, A_{\text{anth}}, \text{surface}, \text{illumination}, \text{camera}, \text{zone}) + \epsilon.$$

The observation vector can include L, a, b*, hue, chroma, and calibrated spectral ratios. The function g may be linear/PLS, monotone spline, or a small learned observation network. Chemical pigment assays are required to interpret states as concentrations.

Published Hass work supports early chlorophyll decline and later cyanidin-3-O-glucoside accumulation ([Cox et al.](#)). Temperature-related darkening can desynchronize colour and internal readiness ([Sibeko et al.](#)).

E.9 Carbohydrate and soluble-solid balance

$$\frac{dS_{\text{starch}}}{dt} = -k_S(T) S_{\text{starch}},$$

$$\frac{dS_{\text{sol}}}{dt} = Y_S k_S(T) S_{\text{starch}} - r_{\text{resp, sol}} - r_{\text{other}}.$$

Approximate measured concentration:

$$C_{\text{sol}} = \frac{S_{\text{sol}}}{M_w S_{\text{sol}} + M_{\text{other, sol}}}.$$

Interpretation: Brix is a concentration influenced by carbohydrate conversion/consumption and water change. It cannot independently identify starch conversion.

Required observations: moisture or water mass, Brix, and a validated starch/carbohydrate assay on matched tissue. Sampling location matters because the fruit may be heterogeneous.

E.10 Lipid and fresh-basis concentration

Most avocado lipid deposition occurs before harvest. A conservative postharvest candidate is:

$$\frac{dM_L}{dt} = -r_L(T, R), \quad C_{L, \text{fresh}} = \frac{M_L}{M}.$$

A rise in $C_{L, \text{fresh}}$ can occur when water/total mass falls, even if M_L is constant. Do not insert a postharvest "oil accumulation" term without independent evidence. Oil extraction or a validated reference method is needed for calibration. Broad NIR alone does not guarantee oil specificity because water and scattering overlap.

E.11 Latent ripeness models

E.11.1 Logistic state

$$\frac{dR}{dt} = k_R(T, RH, E)R(1-R).$$

E.11.2 Gompertz state

$$R(t) = \exp[-\exp\{-k_R(t - t_0)\}].$$

E.11.3 Multistate alternative

A single R may be inadequate. A minimal vector state could be:

$$\mathbf{x} = [F, M_w, C_{\text{chl}}, A_{\text{anth}}, E, r_C]^T,$$

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, T, RH, v; \boldsymbol{\theta}_p) + \boldsymbol{\eta},$$

with fruit-specific parameters

$$\log \boldsymbol{\theta}_p = \boldsymbol{\mu} + \mathbf{B}_{\text{batch}} + \mathbf{b}_{\text{fruit}}.$$

This nonlinear mixed-effects construction separates population, batch, and fruit variation. It should be compared with the one-state model on held-out batches and posterior predictive checks.

E.12 Sensor observation model

For sensor channel i :

$$y_{i,t} = h_i(\mathbf{x}_t, \mathbf{u}_t, \text{device}_i, \text{zone}) + b_i(t) + \epsilon_{i,t},$$

where $\mathbf{u}_t$ contains environment and protocol inputs, $b_i(t)$ is drift, and $\epsilon_{i,t}$ is measurement noise.

Examples:

- RGB depends on pigments, surface state, illumination, camera, and zone.
- 970-nm response depends on water absorption, scattering, geometry, temperature, and detector/source behavior.
- scale mass depends on fruit state, tare, temperature, creep, and off-axis load.
- MOX resistance depends on mixed volatiles, RH, T, baseline, and heater history.
- chamber gas depends on production, volume, mixing, leaks, adsorption, and sensor dynamics.

Calibrations and blank/reference observations belong in the likelihood, not only in a preprocessing spreadsheet.

E.13 Remaining-useful-life formulation

Define a prespecified acceptable region $\mathcal{A}$ over firmness, internal condition, and any required sensory endpoint. At observation time t :

$$\mathrm{RUL}(t) = \inf_{\tau > 0} \{ \tau : \mathbf{x}(t + \tau) \notin \mathcal{A} \}.$$

The entry and exit times are interval-censored when fruit is inspected only at sessions. For destructive states, the sentinel fruit endpoint is latent and informed by matched sacrifice cohorts. Forecasts should be distributions:

$$p(\mathrm{RUL}(t) \mid \mathbf{y}_{1:t}, \mathbf{u}_{1:t}, \mathcal{D}_{\mathrm{train}}),$$

not a single number without uncertainty.

E.14 Estimation methods

Method	Recommended use	Required safeguard
Nonlinear least squares	Simple single-fruit/mean trajectory diagnostics	Cluster-aware uncertainty and residual checks
Nonlinear mixed effects	Population rates plus fruit/batch variability	Structural identifiability and random-effect diagnostics
Bayesian state-space/MCMC	Latent states, missing data, full uncertainty	Prior sensitivity, convergence, posterior predictive checks
Extended/unscented Kalman filter	Sequential online update	Test nonlinearity and non-Gaussian failure
Particle filter	Nonlinear/non-Gaussian sequential inference	Effective sample size and resampling diagnostics
Gaussian process	Small-data trajectory smoothing and uncertainty	Avoid derivative overconfidence
SINDy	Sparse candidate rate terms	Smooth/noise study, units, stability, external refit
Symbolic regression	Alternative human-readable functions	Constrained search, dimensional consistency, held-out confirmation
Neural ODE	Flexible irregular-time latent dynamics	Independent trajectories, regularization, interpretability boundary
Universal differential equation	Known balance plus learned residual	Ensure residual does not silently absorb confounding
PINN	Defensible ODE/PDE with boundary information	Optimization diagnostics and comparison with classical solvers

E.15 Identifiability workflow

1. Draw the causal/measurement graph and list measured, latent, controlled, and unobserved variables.
2. Nondimensionalize or enforce units.
3. Perform structural identifiability analysis for each proposed equation set.
4. Use synthetic parameter-recovery tests only as **pipeline verification**.
5. Fit the simplest candidate to pilot data.
6. Inspect profile likelihood or posterior geometry, parameter correlations, residuals, and state uncertainty.
7. Add one coupling only when the relevant state is independently observed.
8. compare candidates on leave-one-batch/condition-out prediction;
9. test intervention prediction at withheld T/RH;
10. reject or simplify models with nonphysical states, weak recovery, wide parameters, or failed external forecasts.

E.16 Candidate initial/boundary conditions

Process	Initial condition	Boundary/forcing information
Mass/water	First stable mass; matched initial moisture distribution	T, RH, airflow, area; skin convective boundary if PDE
Firmness	Matched batch/initial destructive distribution	Temperature and optional ethylene state
Ethylene	Post-flush chamber baseline and latent basal production	Seal/vent schedule, free volume, leaks, adsorption
CO ₂	Ambient/empty-chamber baseline	Seal time, volume, pressure, T, leaks, sensor response
Pigments	Initial calibrated colour plus chemical subset	Illumination/device/zone observation model
Carbohydrate	Initial starch/Brix/moisture distribution	Respiration/carbohydrate consumption assumptions
Latent R	Batch-informed distribution, not arbitrary zero for all fruit	Recorded T/RH/handling history

E.17 Model-selection decision rule

A dynamic model advances only if it:

1. passes numerical and unit tests;
2. produces physically admissible states;
3. has identifiable or decision-useful parameter combinations;
4. improves batch-held-out forecast error, calibration, interval coverage, or temporal consistency over a simpler baseline;
5. remains stable under plausible initial-condition and measurement-error perturbations;
6. reports uncertainty and abstains outside its validated domain.

A visually plausible fitted curve is not sufficient evidence of a governing law.

Appendix F - Proposed Software Stack

Project: Multimodal non-destructive Hass avocado ripeness characterisation **Status:** reproducible-stack design **Version policy:** no runtime or package version has been verified locally for this appendix; every version below is therefore marked to be pinned

F.1 Reproducibility policy

The software stack should be fixed only after the target workstations, embedded boards, and any university high-performance-computing environment are confirmed. "Latest" must never appear in an environment definition. The implementation should:

1. record operating system, architecture, compiler, firmware, interpreter, package, driver, and instrument-software versions;
2. pin direct and transitive dependencies in lock files;
3. preserve raw data separately from calibrated and derived outputs;
4. place every result under a configuration, code commit, data-manifest, calibration, and random-seed identity;
5. execute validation and analysis through scripts or workflow targets rather than undocumented notebook state;
6. export open, archival tables and figures in addition to proprietary formats.

Because versions have not been locally verified, the first implementation task is to generate and review the lock files on the actual acquisition and analysis computers.

F.2 Stack overview

flowchart TD

```

A["ESP32-S3 firmware<br/>sensor acquisition and sequencing"] --> B["Raw CSV/JSONL/binary records<br/>microSD + serial transfer"]
C["Raspberry Pi camera service<br/>locked image settings"] --> D["Images + capture metadata"]
E["Spectrometer / gas-analyser vendor export"] --> F["Immutable instrument-native files<br/>plus open export"]
B --> G["Ingestion and schema validation"]
D --> G
F --> G
G --> H["Calibrated data layer"]
H --> I["Exploratory and statistical analysis"]
H --> J["Classical machine learning"]
H --> K["Time-series and dynamical modelling"]
H --> L["Scientific machine learning"]
I --> M["Figures, tables, reports"]
J --> M
K --> M
L --> M
N["KiCad + CAD + drawings"] --> O["Hardware release package"]
P["Git + CI + environment locks"] --> M
P --> O

```

F.3 Firmware and embedded acquisition

Component	Proposed choice	Version	Purpose
ESP32 framework	Espressif ESP-IDF	To be pinned	Production firmware, drivers, tasks, storage, communications, and hardware test
Optional development layer	PlatformIO Core and Espressif32 platform	To be pinned	Repeatable local builds and board configuration if it does not obscure ESP-IDF versions
Language/toolchain	C/C++ and Espressif GCC toolchain	To be pinned	Deterministic embedded implementation
Build system	CMake/Ninja as supplied by the chosen ESP-IDF release	To be pinned	Reproducible firmware build
Unit testing	Unity test framework supplied by ESP-IDF	To be pinned with ESP-IDF	Sensor-driver and conversion tests
Static analysis	clang-format, clang-tidy, cppcheck	To be pinned	Formatting and defect checks
Serial protocol	Versioned newline-delimited JSON for development; compact framed binary or CSV where required	Schema version to be defined	Observable transfer and recovery
Local storage	FAT-compatible microSD with append-only session files	Format/procedure to be frozen	Power-loss-tolerant raw logging

Firmware responsibilities:

- hardware and firmware identity at boot;
- monotonic and UTC timestamps with synchronisation status;
- non-blocking schedules for SHT45, SCD30, BME688, HX711, ADS1115, fan, pump, and LEDs;
- explicit chamber state machine: open/flush, equilibrate, accumulate, vent, optical, weigh;
- LED wavelength, current, detector gain, integration, dark/reference/fruit state, and saturation flags;
- raw sensor counts alongside converted engineering units;
- watchdog, brownout, SD-write, sensor-timeout, range, and CRC/error records;
- a safe output state after reboot or communication loss;
- a hardware-in-the-loop diagnostic mode.

Suggested firmware repository location:

```
firmware/
  CMakeLists.txt
  sdkconfig.defaults
  partitions.csv
main/
  components/
    acquisition/
    chamber_state/
    sensors/
    optical_head/
    storage/
    telemetry/
  test/
  tools/
```

Do not commit generated build directories or credentials. Commit `sdkconfig.defaults`, pin the ESP-IDF release, and archive the full build manifest with each flashed binary.

F.4 Camera acquisition

Component	Proposed choice	Version	Purpose
Raspberry Pi OS Lite	64-bit image compatible with Zero 2 W	To be pinned by image date/hash	Minimal camera host
Camera API	libcamera/rpicam applications and Picamera2	To be pinned together	Manual camera control and metadata
Camera service	Python service using Picamera2 or a shell-free native wrapper	To be pinned	Session-triggered image acquisition
Image processing	OpenCV-Python, Pillow, colour-science	To be pinned	Calibration target detection and colour extraction
Metadata	ExifTool or native sidecar JSON writer	To be pinned	Exposure, gain, focus, white balance, device, and session provenance

Camera acquisition must freeze:

- sensor mode and image dimensions;
- focus position;
- exposure time and analogue/digital gain;
- white balance gains;
- denoise, sharpening, HDR, gamma, and compression policy;
- camera-fruit distance, illumination, crop, and orientation;
- ColorChecker identity and calibration-frame timing.

Retain original captures and JSON sidecars. Produce calibrated L^* , a^* , b^* , hue, chroma, patch residuals, fruit masks, and quality flags in the derived layer; do not overwrite the originals.

F.5 Instrument acquisition

Vendor applications may be unavoidable for the spectrometer, F-900, moisture analyser, or other laboratory instruments. The acquisition policy is:

1. retain the untouched native file;
2. export an open CSV or text representation without deleting the native file;
3. record instrument model, serial, firmware, vendor-software version, method, integration time, averaging, dark/white reference, calibration file, flow, and operator;
4. compute a SHA-256 hash of every raw file after transfer;
5. map each observation to `study_id`, `fruit_id`, `session_id`, scan zone, azimuth, UTC time, and calibration ID.

Where an instrument offers an SDK:

Instrument layer	Proposed interface	Version
Ocean Insight spectrometer	Manufacturer-supported API/SDK or SeaBreeze-compatible interface validated for the purchased model	To be pinned after model quotation
Hamamatsu microspectrometer	Project-specific ESP32/desktop readout designed from the exact datasheet timing requirements	Firmware and acquisition package to be pinned
Felix F-900	Manufacturer export/software and documented serial or file interface	To be pinned after instrument access
Laboratory balance/moisture equipment	Vendor export or validated manual double-entry workflow	To be pinned or method-controlled

An unsupported community driver should not become the only way to recover primary instrument data.

F.6 Core Python analysis environment

Python is the principal analysis language. MATLAB and R remain optional where a university method or collaborator requires them.

Package/tool	Version	Role
Python	To be pinned	Analysis runtime
uv or Conda/Mamba	To be selected and pinned	Environment and lock management
NumPy	To be pinned	Numerical arrays and linear algebra
SciPy	To be pinned	Optimisation, integration, signal processing, statistics
pandas	To be pinned	Tabular and longitudinal data handling
Polars	Optional; to be pinned	Fast columnar processing for large spectral tables
PyArrow	To be pinned	Parquet/Arrow interchange
h5py and/or Zarr	To be pinned	Dense spectral/time-series arrays where Parquet is unsuitable
xarray	To be pinned	Labelled wavelength/time arrays
pydantic	To be pinned	Configuration and record-schema validation
pandera	To be pinned	Data-frame schema validation
pint	To be pinned	Unit-aware calculations
OpenCV	To be pinned	Image registration, masks, target detection, colour extraction
Pillow	To be pinned	Image I/O and metadata support
colour-science	To be pinned	Colour-space conversion and colour difference
scikit-image	To be pinned	Image features and quality checks
matplotlib	To be pinned	Static publication figures
seaborn	To be pinned	Statistical plots
Plotly	Optional; to be pinned	Interactive diagnostics
JupyterLab	To be pinned	Exploration only; final outputs must run from scripts/workflows

Recommended open formats:

- CSV for small human-inspectable acquisition records;
- Parquet for validated tabular observations and features;
- TIFF/PNG or documented original camera format plus JSON sidecars for images;
- HDF5 or Zarr for large wavelength × sample arrays;
- YAML or TOML for human-authored configuration;
- JSON for immutable run manifests;
- SVG/PDF/PNG for figures;
- Markdown, BibTeX, CSL JSON, and PDF for documentation.

F.7 Quality control, signal processing, and chemometrics

Package/tool	Version	Role
SciPy signal module	To be pinned with SciPy	Filters, peak checks, numerical derivatives
pybaselines	To be pinned	Baseline correction where justified
pywavelets	Optional; to be pinned	Wavelet denoising experiments
scikit-learn	To be pinned	Preprocessing, PLS, PCA, cross-validation, metrics
statsmodels	To be pinned	Regression, repeated-measures and diagnostic models
pingouin	Optional; to be pinned	Convenient statistical summaries after method review
lmfit	To be pinned	Constrained nonlinear parameter fitting
ruptures	Optional; to be pinned	Change-point analysis

Candidate spectral preprocessing should be treated as a model hyperparameter inside grouped validation, not chosen after inspecting test outcomes. Candidates include:

- dark subtraction and reference normalisation;
- absorbance transform where physically appropriate;
- standard normal variate;
- multiplicative scatter correction;
- Savitzky-Golay smoothing and first/second derivatives;
- wavelength selection;
- replicate aggregation with retained within-fruit variance.

Every transform must record its fitted parameters and be learned from training data only.

F.8 Statistical analysis

Component	Proposed choice	Version
Classical and mixed-effects statistics	statsmodels	To be pinned
Optional R workflow	R, lme4, nlme, emmeans, brms	To be pinned only if adopted
Bayesian modelling	PyMC and ArviZ	To be pinned
Alternative probabilistic stack	Stan via CmdStanPy	Optional; to be pinned
Power/simulation	Python simulation scripts; optional R packages	To be pinned

Fruit ID, orchard/batch, storage block, acquisition day, scan zone, and repeated measurements should be represented explicitly. Confidence intervals and uncertainty must reflect biological replication, not the number of spectra alone.

F.9 Machine-learning stack

F.9.1 Baseline and tabular models

Tool	Version	Intended models
scikit-learn	To be pinned	PLS regression, logistic/ordinal regression, SVM, random forest, elastic net, Gaussian process, PCA
XGBoost	To be pinned	Gradient-boosted trees
LightGBM	To be pinned	Gradient-boosted trees where installation/support is acceptable
Optuna	To be pinned	Nested hyperparameter optimisation
imbalanced-learn	Optional; to be pinned	Training-fold-only imbalance handling
SHAP	To be pinned	Post-hoc feature attribution with stated limitations

PLS, regularised linear models, and tree ensembles are the primary baselines for modest biological sample sizes. Deep learning should not precede strong grouped baselines.

F.9.2 Deep and multimodal learning

Tool	Version	Intended use
PyTorch	To be pinned	Principal neural-network framework
torchvision	To be pinned with PyTorch	RGB encoders and image transforms
Lightning	Optional; to be pinned	Structured training loops and checkpointing
TensorFlow/Keras	Optional alternative; to be pinned only if selected	Do not maintain duplicate production implementations
JAX	Optional; to be pinned only if SciML method requires it	Differentiable numerical models
einops	To be pinned if used	Auditable tensor rearrangement
timm	Optional; to be pinned	Pretrained image encoders

Candidate architectures include one-dimensional spectral CNNs, temporal convolutional networks, LSTM/GRU models, small transformers, and late-fusion multimodal networks. Model size must be constrained by the number of independent fruit and biological batches.

F.10 Scientific machine learning and dynamical models

Method	Proposed software	Version	Role
Candidate ODE/PDE fitting	SciPy <code>solve_ivp</code> , optimisation, and least squares	To be pinned	Transparent mechanistic baselines
Neural ODE	<code>torchdiffeq</code> or equivalent	To be pinned if selected	Learn continuous latent ripening dynamics
Sparse equation discovery	PySINDy	To be pinned	Candidate governing-term discovery
Physics-informed networks	DeepXDE or a reviewed PyTorch/JAX implementation	To be pinned if selected	Soft physical constraints and inverse problems
Differentiable ODE solvers	DiffraX if JAX is selected	To be pinned if selected	Neural/differentiable dynamical models
Symbolic regression	PySR	To be pinned	Interpretable candidate rate laws
State-space estimation	<code>filterpy</code> or custom reviewed implementation	To be pinned if selected	Kalman/extended/unscented filtering
Bayesian dynamical inference	PyMC or Stan	To be pinned	Parameter/posterior uncertainty
Gaussian processes	scikit-learn and/or GPyTorch	To be pinned if selected	Nonlinear trends and calibrated uncertainty

These tools infer or fit **candidate** models. They do not establish a governing equation merely because a low residual is obtained. Evaluation must include dimensional consistency, identifiability, parameter uncertainty, residual structure, extrapolation, held-out fruit/batches, and comparison with simpler kinetics.

F.11 Experiment tracking and validation

Component	Proposed choice	Version
Configuration	Hydra/OmegaConf or typed TOML/YAML with pydantic	To be selected and pinned
Experiment tracking	MLflow, or local immutable run manifests if a service is unnecessary	To be selected and pinned
Data versioning	DVC or lakeFS only if operational support exists; otherwise content-addressed manifests	To be selected and pinned
Tests	pytest, pytest-cov, hypothesis	To be pinned
Lint/format	Ruff	To be pinned
Type checking	mypy or Pyright	To be selected and pinned
Security/dependency audit	pip-audit and repository secret scanning	To be pinned
Workflow orchestration	Snakemake or Make	To be selected and pinned
Containers	Docker/Podman where permitted	Engine and base-image digest to be pinned
Version control	Git	To be pinned/documented by environment

Minimum automated checks:

- schema and unit validation;
- duplicate and impossible timestamp checks;
- calibration linkage and range flags;
- train/test fruit-ID separation;
- deterministic-seed and environment capture;
- unit tests for all conversions and rate calculations;
- synthetic tests for ODE/parameter-recovery code;
- notebook execution from a clean kernel;
- figure/table reproduction from immutable inputs;
- firmware build and host-parser compatibility.

F.12 Data splitting and model-evaluation safeguards

All spectra, zones, images, and days belonging to one fruit must remain in the same outer split. Where the intended deployment crosses harvest batches, orchards, seasons, storage regimens, cameras, or prototypes, those domains should be held out explicitly.

Recommended hierarchy:

1. development set with grouped cross-validation by fruit and batch;
2. untouched internal test set of fruit;
3. external batch/season/device validation;
4. prospective confirmation after model freeze.

Hyperparameter optimisation, preprocessing, imputation, feature selection, augmentation, calibration, and threshold selection must occur inside the training folds. Report fruit-level and observation-level metrics separately, including uncertainty.

F.13 CAD, enclosure, PCB, and engineering documentation

Discipline	Proposed tool	Version	Output
PCB/schematic	KiCad	To be pinned	Source project, schematic PDF, Gerbers, drill, BOM, pick-and-place, ERC/DRC reports
Parametric mechanical CAD	FreeCAD and/or OpenSCAD	To be selected and pinned	Editable open-source geometry and STEP/STL exports
Institutional/commercial CAD	Fusion 360 or SolidWorks where licences exist	To be pinned if used	Native source plus STEP/DXF/PDF neutral exports
Diagramming	Mermaid and SVG source	Renderer to be pinned	Architecture and protocol diagrams
Electronics simulation	LTspice or ngspice	To be selected and pinned	TIA and driver simulation source/results
3D slicing	PrusaSlicer or Cura	To be pinned if printed parts are used	Project file, material/profile, G-code metadata

Mechanical and PCB releases should carry a revision, compatible BOM revision, drawing dimensions/tolerances, material and surface-finish specification, and assembly/calibration instructions. Gas-wetted materials and optical black coatings require explicit blank/recovery validation.

F.14 Documentation, references, and publication

Tool	Version	Role
Markdown	CommonMark/GitHub-compatible source	Human-readable canonical documentation
Pandoc	To be pinned	Markdown-to-PDF/DOCX conversion
LaTeX engine	TeX Live and selected engine	To be pinned by distribution/date
Quarto	Optional; to be pinned	Reproducible manuscripts/reports if adopted
Zotero	To be pinned/documented	Reference library
Better BibTeX	To be pinned with Zotero	Stable citation keys and BibTeX export
CSL style	Commit exact .cs1 file	Journal/funder citation format
Graphviz/Mermaid CLI	To be pinned	Deterministic diagram rendering

The appendices link to the project's local source records:

- Prototype architecture
- Audited BOM
- Detailed hardware catalogue
- Procurement source ledger
- Hardware Appendix D

For PDF production, the build should resolve these relative links from the repository root or copy linked assets into a controlled staging directory without altering their names or hashes.

F.15 Proposed repository structure

```

.
├── analysis/
│   ├── configs/
│   ├── notebooks/
│   ├── src/
│   ├── tests/
│   └── workflows/
├── appendices/
├── cad/
│   ├── chamber/
│   ├── fixtures/
│   └── exports/
├── data/
│   ├── external/
│   ├── manifests/
│   ├── processed/
│   ├── raw/
│   └── schemas/
├── docs/
├── firmware/
├── ml/
│   ├── configs/
│   ├── models/
│   ├── tests/
│   └── training/
├── models/
│   ├── bayesian/
│   ├── candidate_odes/
│   ├── pinn/
│   ├── sindy/
│   └── symbolic/
├── pcb/
├── procurement/
├── proposal/
├── references/
├── reports/
├── results/
│   ├── figures/
│   ├── metrics/
│   ├── models/
│   └── tables/
└── scripts/

```

Raw and external datasets should normally be excluded from Git and governed by checksummed manifests, licence terms, and access controls. Small schemas, metadata, calibration templates, and synthetic test fixtures should be version-controlled.

F.16 Environment artefacts to create before analysis

The implementation phase should produce:

- `firmware/esp-idf-version.txt` or equivalent exact framework commit;
- `firmware/sdkconfig.defaults`;
- `pyproject.toml`;
- an exact Python lock file generated on the target platform;
- optional Conda environment lock for native scientific dependencies;
- container definition with base-image digest where containers are allowed;
- `CITATION.cff`;
- `LICENSE` and third-party notices;
- pre-commit configuration;
- CI workflow;
- schema and unit registry;

- dataset manifest specification;
- model/run manifest specification;
- pinned Pandoc/LaTeX/diagram build environment.

An environment is accepted only after a clean machine can acquire a synthetic/device test session, validate the records, run the analysis test suite, train a small baseline, and reproduce a reference figure and report without undocumented manual steps.

F.17 Security, privacy, and operational constraints

- Do not place Wi-Fi passwords, API tokens, participant information, or supplier credentials in the repository.
- Use least-privilege device/network access; acquisition should continue locally during network failure.
- Validate all imported CSV, image, instrument, and configuration files.
- Archive checksums before transforming raw instrument exports.
- Record destructive edits and exclusions in an auditable data-quality table.
- Back up raw data to at least two independent media/locations under institutional policy.
- Treat firmware update, time synchronisation, and device replacement as controlled changes that can shift the measurement domain.

F.18 Final selection gate

Before pinning the production stack, the team should confirm:

1. host operating systems and CPU/GPU architectures;
2. exact ESP32 board revision and camera computer;
3. purchased/shared spectrometer and its supported SDK;
4. F-900/GC and laboratory export formats;
5. university restrictions on Docker, cloud tracking, MATLAB, Fusion 360, and data storage;
6. publication target and required citation/report format;
7. whether JAX, TensorFlow, R, MATLAB, or commercial CAD adds a method not already covered by the principal stack.

The preferred minimal production path is **ESP-IDF + Raspberry Pi camera tools + Python/SciPy/pandas/scikit-learn/PyTorch + PyMC/PySINDy as justified + KiCad + open neutral CAD exports + Git and locked environments**. Optional frameworks should be added only for a defined scientific requirement, not as parallel implementations of the same workflow.

Appendix G - Repository Structure

G.1 Implemented workspace

```

Avocado/
  analysis/
    inspect_downloads.py
    prepare_rgb_dataset.py
    run_eda.py
    draw_prototype_overview.py
  appendices/
    appendix-a-existing-papers.md
    ...
    appendix-l-risk-evidence.md
  communications/
    collaborator-outreach-draft.md
  data/
    dataset_inventory.csv
    raw/                immutable source archives and inspected repositories
    processed/           reproducibly derived metadata and RGB features
  docs/
    dataset-audit.md
    data-dictionary.md
    prototype-architecture.md
  figures/               generated EDA, ML, and architecture figures
  ml/
    train_grouped_baselines.py
    artifacts/           reproducibility artifacts, not deployment models
    results/             folds, splits, OOF predictions, machine summaries
  models/
    kinetic/             candidate ODE scaffold and documentation
    sciml/               scientific-ML implementation plan
  Papers(toReadAndCiteMaybeLater)/
    user-supplied source PDFs
  procurement/
    bill-of-materials.csv
    equipment-catalogue.md
    source-ledger.md
    images/              licence-verified exact-product photographs
  proposal/
    avocado-research-proposal.md
  reports/               QA, EDA, and ML narrative reports
  reproducibility/
    requirements-pinned.txt
    source-checksums.csv
    validate_outputs.py
    run_all.ps1
  tables/                publication-ready machine-readable tables
  output/pdf/            final rendered deliverable only
  tmp/                   disposable extraction, render, and support files

```

G.2 Future experimental extensions

```

firmware/
  esp32/                sensor acquisition, LED control, SD logging
  raspberry-pi/         locked camera acquisition and session trigger
pcb/
  kicad/                schematics, layout, BOM, fabrication outputs
cad/
  chamber/
  optical-fixture/
  fruit-cradle/
data/
  raw/
    study_id/device_id/date/session_id/
  calibration/
  interim/
  processed/
protocols/
  metrology/
  biological/
  destructive-assays/
  safety/
schemas/
  session.schema.json
  sensor-record.schema.json
  destructive-record.schema.json
models/
  chemometrics/
  temporal/
  fusion/
  kinetic/
  sciml/
tests/
  firmware/
  schema/
  analysis/
  model-splits/
model-cards/
preregistration/

```

G.3 Data-layer contract

1. **Raw:** byte-preserved device/source output. Never edited.
2. **Calibration:** dark, white, colour, mass, gas, wavelength, blank, leak, and response records.
3. **Interim:** decoded but not scientifically transformed records.
4. **Processed:** validated, calibrated observations linked to immutable raw IDs and code/calibration versions.
5. **Features:** fold-safe derived predictors.
6. **Results:** split manifests, predictions, metrics, uncertainty, and diagnostics.

Every derived file must be reproducible from a lower layer. Destructive samples retain fruit, session, zone, method, aliquot, operator, and calibration keys.

G.4 Naming and identity rules

- `fruit_id` identifies the biological unit and never changes.
- `batch_id`, `chamber_id`, `device_id`, and `calibration_id` are explicit.
- filenames do not serve as hidden predictor labels;
- patches, sides, days, and augmented records retain their parent image/fruit;
- timestamps are stored in UTC with local context retained separately;

- units appear in the schema and metadata, not only in a notebook;
- raw inputs and locked test cohorts are read-only.

G.5 Version-control and release rules

- Source code, schemas, small tables, documentation, calibration metadata, and manifests belong in version control.
- Large raw datasets and model artifacts use an institutional object store, data registry, or approved large-file mechanism; checksums remain in the repository.
- Dataset licensing is evaluated independently from article and code licensing.
- Secrets, personal data, supplier credentials, and unpublished participant information are excluded.
- A release includes environment lock, data/source manifest, split manifest, model card, known limitations, and exact regeneration command.

G.6 Recommended branches/tags

- protected main for validated releases;
- short-lived feature branches for analysis, firmware, PCB, and protocol work;
- signed or annotated tags for protocol-v1, data-lock-v1, model-freeze-v1, and manuscript releases;
- a decision record for every post-lock change.

Appendix H - Data Quality and Data Dictionary

H.1 Primary dataset

The audited primary source is Xavier, Rodrigues, and Silva, *'Hass' Avocado Ripening Photographic Dataset*, version 1 (2024), DOI [10.17632/3xd9n945v8.1](https://doi.org/10.17632/3xd9n945v8.1), CC BY 4.0.

The immutable primary archive is `data/raw/Hass-Avocado-Ripening-Photographic-Dataset-v1.zip`.

Bytes: 418384370
SHA-256: FB14C3D8C6FB59A20BFFB579A5FE97C7EBA7030F4420882E8A75B575FF0C5BB6

H.2 Reconciliation results

DATA-DEMONSTRATED:

Check	Result
Workbook records	14,722
JPEG files	14,710
Exact filename matches	14,710
Workbook rows missing an image	12
Images absent from workbook	0
Duplicate workbook filenames	0
Independent biological fruits	478
Filename/workbook metadata disagreements	0
Within-fruit, within-side stage reversals	0

No missing JPEG was synthesized, imputed, or replaced. The reconciled metadata table preserves all 14,722 workbook records with `image_present`; the feature table contains only the 14,710 records with an available image.

H.3 Biological and technical hierarchy

```
study
├── storage group
│   ├── biological fruit
│   │   ├── day/session
│   │   │   ├── side/view
│   │   │   │   └── image pixels and derived features
```

The biological fruit is the minimum validation group. Image side, repeated day, pixel, patch, or augmented copy is not an independent fruit. The defensive project identifier is `fruit_id = storage_group + sample_id`.

H.4 Core source and derived variables

Variable	Unit/domain	Provenance	Modeling rule
file_name	string	Source workbook	Row key; never a predictor
timestamp	ISO datetime	Source workbook	Audit only
storage_group	T10/T20/Tam	Source workbook	Stress-test/covariate; excluded from image-only baseline
sample_id	1-478	Source workbook	Biological group; never a predictor
fruit_id	string	Derived	Mandatory split group
day	integer days	Source workbook	Longitudinal index; excluded from image-only baseline
side	a/b	Filename parse	Repeated view; excluded from predictor matrix
ripening_stage	ordered 1-5	Source study composite index	Classification target
image_present	Boolean	File reconciliation	Quality flag
first_stage4_day	day/blank	First observed stage ≥ 4 by fruit	Dataset-specific derived endpoint
days_to_stage4_signed	days/blank	endpoint day minus record day	Retrospective target, not independent shelf-life truth
RGB/HSV/gray/edge features	numeric	Reproducible image extraction	Predictor candidates

The complete column-by-column specification, units, derivations, and interpretation boundaries are maintained in ``docs/data-dictionary.md``.

H.5 Endpoint limitations

The five-level ripening index combines external appearance and manual texture assessment. It is not instrumental firmness, dry matter, moisture, oil, ethylene, internal-condition, or sensory acceptability. A model trained on it estimates the source study's operational index.

The first stage-4 day is the first *observed* record at that stage, not a continuously observed transition. Fifty-two fruits did not reach stage 4 and 68 did not reach stage 5 during follow-up. A future event model must treat these as censored observations rather than assigning arbitrary endpoint times.

H.6 Quality gates for new experimental data

1. Validate identifiers, units, ranges, monotonic clocks, calibration IDs, and parent-child keys on ingestion.
2. Preserve raw images, spectra, and sensor counts unchanged.
3. Store calibration transforms and derived values in separate versioned tables.
4. Record missingness and exclusion reasons rather than deleting records.
5. Detect duplicate and near-duplicate images before splitting.
6. Assert zero fruit overlap in every model split.
7. Keep entire harvest batches and the locked external cohort outside model selection.
8. Link every destructive aliquot to fruit, zone, time, method, operator, and non-destructive history.

Machine-readable reconciliation results are in ``reports/data-quality-summary.json``; the narrative audit is ``reports/data-quality-report.md``.

Appendix I - Machine-Learning Results

I.1 Evidence boundary

The completed models use only the CC BY 4.0 Xavier et al. RGB dataset and foreground-aware colour/texture summaries. Results are **DATA-DEMONSTRATED internal validation** for 478 fruits in one source study. They are not an external-device, independent-season, chemistry, or deployment validation.

Filename, fruit identity, elapsed day, image side, storage condition, and targets were excluded from predictors. Every observation from one fruit stayed within one fold. The split audit reports zero fruit overlap.

I.2 Five-stage classification

Five-fold stratified, fruit-grouped out-of-fold results:

Model	Accuracy	Balanced accuracy	Macro F1	Ordinal MAE
Prior-only dummy	0.243	0.200	0.078	1.977
Multinomial logistic	0.704	0.699	0.698	0.321
Random forest	0.737	0.726	0.727	0.280
Histogram gradient boosting	0.739	0.731	0.730	0.278

Fruit-cluster bootstrap 95% intervals for the selected model were:

- accuracy 0.727-0.751;
- balanced accuracy 0.719-0.743;
- macro F1 0.719-0.742;
- ordinal MAE 0.264-0.293.

I.3 Storage-condition holdout

Held-out group	Test fruits	Accuracy	Balanced accuracy	Macro F1
10 C / 85% RH	192	0.478	0.497	0.463
20 C / 85% RH	143	0.680	0.688	0.677
Ambient	143	0.618	0.632	0.608

The low 10 C result demonstrates material condition shift. It is stronger evidence for the need for multimodal and external validation than a single pooled score is evidence for deployment.

I.4 Days to first observed stage 4

The retrospective target includes only observations on or before the first record at stage 4 for the 426 fruits that reached stage 4. It contains 8,834 images and is derived from the same ordinal labels.

Model	MAE (days)	RMSE (days)	R-squared
Median dummy	3.746	4.754	-0.034
Ridge	2.158	2.978	0.595
Random forest	1.881	2.656	0.677
Histogram gradient boosting	1.853	2.636	0.682

The selected model's fruit-cluster bootstrap 95% intervals were 1.783-1.922 days for MAE, 2.542-2.727 days for RMSE, and 0.655-0.704 for R-squared.

I.5 Most influential features

Permutation importance ranked red-to-green ratio, excess green, green chromaticity, green-to-blue ratio, and foreground hue summaries highest. These are predictive associations and do not establish that peel colour uniquely determines internal readiness.

I.6 Reproducibility map

- Narrative with full intervals and interpretation:

``reports/ml-results.md``

- Training code: ``ml/train_grouped_baselines.py``
- Model table: ``tables/model-comparison.csv``
- Storage stress test:

``tables/storage-holdout-results.csv``

- Split audit: ``ml/results/split-audit.csv``
- Fold metrics: ``ml/results/fold-metrics.csv``
- All out-of-fold predictions:

``ml/results/oof-predictions.csv``

- Machine-readable summary:

``ml/results/summary.json``

- Full-data comparison artifact:

`ml/artifacts/final_color_stage_model.joblib` (not deployment qualified)

I.7 Required next validation

No model should be advanced to use until it is tested on independent harvest batches and a locked season/device cohort, with instrumental firmness, internal-condition, and a prespecified remaining-useful-life endpoint. The storage holdout should be repeated for the multimodal model. Calibration, subgroup error, prediction interval coverage, and abstention under domain shift are mandatory.

Appendix J - Statistical Analysis Plan

J.1 Status

This is a **PROPOSED** preregistration scaffold. Endpoint thresholds, smallest useful effects, final sample size, and confirmatory model degrees of freedom must be locked after the metrology/pilot stages and before confirmatory outcomes are inspected.

J.2 Experimental units and hierarchy

- biological unit: individual avocado;
- repeated technical/longitudinal units: scan zone, side, image, spectrum, and session;
- higher units: storage chamber, harvest/receipt batch, orchard/season, and device;
- destructive aliquots: nested within a sacrificed fruit and zone.

Images, spectra, patches, wavelengths, and days do not increase the independent fruit count. A condition effect requires independently replicated chambers or a design that rotates conditions across chambers without carry-over.

J.3 Primary estimands

1. Difference in locked-batch firmness MAE between the full multimodal model and calibrated RGB-only baseline.
2. Difference in locked-batch remaining-useful-life MAE between a longitudinal multimodal model and a current-observation model.
3. Coverage of nominal 90% or 95% prediction intervals for those targets.

The stakeholder-defined smallest useful differences and non-inferiority margins are entered before confirmatory enrollment.

J.4 Secondary estimands

- temperature, RH, and interaction effects on softening, mass-loss, pigment, and respiration rate parameters;
- external RMSEP/bias/RPD for moisture, dry matter, and oil calibration;
- sensor-channel incremental value;
- entry/exit event calibration and time-dependent Brier score;
- method agreement for low-cost versus reference gas channels;
- subgroup performance by batch, device, initial maturity, mass, and internal condition.

J.5 Planning design and power

The working confirmatory layout is 360 fruit:

3 temperatures x 2 RH levels x 3 independent batches x 20 fruit

Within each condition-by-batch cell, eight fruit are longitudinal sentinels and 12 are scheduled sacrifices, three at each of four windows. The final design is chosen by simulation from pilot variance components, serial correlation, chamber/batch

intraclass correlation, censoring, endpoint incidence, and expected assay loss. Power is reported for the independent-unit hierarchy, not for the number of images.

J.6 Allocation, masking, and protocol deviations

Within batch, stratify on initial mass and a prespecified initial maturity proxy, then randomize chamber, position, cohort role, and destructive window. Randomize measurement order. Destructive and image-condition assessors should be blinded to model predictions. Record deviations before opening outcome labels.

J.7 Models

Repeated continuous outcomes

Use hierarchical mixed-effects or Bayesian state-space models with fixed temperature, RH, time, and interactions; random batch, chamber, and fruit effects; and residual correlation/heteroscedasticity selected from pilot data.

Destructive outcomes

Use condition/time fixed effects and batch/chamber random effects. Aliquots are not treated as independent fruit. Report adjusted contrasts and marginal means.

Interval-censored events

Use interval-censored survival with batch/chamber frailty. Fruits not reaching an endpoint are censored. Internal-defect failure can be a competing event or a separate endpoint depending on the signed endpoint definition.

Prediction models

Use nested fruit-grouped development and batch/season/device outer evaluation. All preprocessing, imputation, feature selection, calibration transfer, and hyperparameter selection occur inside training partitions.

J.8 Metrics

Outcome	Primary metrics	Supporting diagnostics
Firmness/chemistry	MAE, RMSE, bias	R-squared, calibration slope, coverage, RPD
Ordinal stage	Balanced accuracy, macro F1, ordinal MAE	Per-stage recall, weighted kappa
Event time	Integrated Brier score, time MAE	Concordance, calibration by horizon
Gas method agreement	Bias and limits of agreement	LOD/LOQ, drift, response/recovery
Dynamic model	Held-out likelihood/error and coverage	Residual autocorrelation, parameter recovery, physical violations

Confidence intervals use fruit- and batch-aware bootstrap or hierarchical posterior intervals. Image-wise resampling is prohibited.

J.9 Missingness and exclusions

Preserve reason codes for saturation, sensor failure, assay failure, fruit damage, contamination, missed visit, and endpoint not reached. The primary analysis population and technical QC exclusions are fixed before data lock. Sensitivity analyses cover complete cases, model-based missingness or multiple imputation, and plausible informative attrition.

J.10 Multiplicity

Use a short ordered primary family or a hierarchical gate. Secondary channel and biomarker screens use false-discovery-rate control. Report estimates and intervals regardless of threshold crossing. Exploratory equation discovery is clearly separated from confirmatory tests.

J.11 Model freeze and locked test

Before unlocking the final external cohort, archive:

- source and feature schema;
- endpoint code;
- training fruit/batch manifest;
- preprocessing and calibration objects;
- hyperparameters and seed;
- model artifact hash;
- exact metrics and subgroup tables;
- abstention/domain-shift rule.

The locked cohort is evaluated once. Any modification after inspection creates a new development cycle and requires another independent test cohort.

J.12 Reporting

Report the flow of fruits from receipt through analysis, condition/batch/chamber counts, exclusions, missingness, all prespecified outcomes, deviations, negative results, intervals, subgroup uncertainty, and the complete split manifest. Model cards must state cultivar, endpoint, equipment, calibration, population, validation boundary, and prohibited uses.

Appendix K - Reproducibility and Provenance Manifest

K.1 Environment

Locally verified on 30 July 2026:

Component	Version
Python	3.12.13
NumPy	2.5.1
pandas	3.0.5
Pillow	12.3.0
openpyxl	3.1.5
PyArrow	21.0.0
scikit-learn	1.7.1
SciPy	1.16.1
Matplotlib	3.10.5
seaborn	0.13.2
joblib	1.5.1
ReportLab	4.4.9
pdfplumber	0.11.9
pypdf	6.10.0

The exact analysis/PDF package list is `reproducibility/requirements-pinned.txt`. ESP-IDF, PlatformIO, camera libraries, CAD/PCB tools, and instrument SDKs are **to be pinned** when the hardware implementation begins; Appendix F does not pretend an unverified future version is current.

K.2 Acquired-source identities

Source	Bytes	SHA-256 / commit
Primary Hass Mendeley ZIP	418,384,370	F814C3D8C6FB59A20BFFB579A5FE97C7EBA7030F4420882E8A75B575FF0C5BB6
Kamat avocado/strawberry ZIP	2,110,769,149	48B8E6B84DD3C58643F624C12689AF6E81A73E2A4CA8D7EB5BBFA7C507BE4224
DeepHS annotations ZIP	165,730	14275450E362684BC379A5AAF6C845CF82B0F9D5912036B9401D59A4B964A3F3
DeepHS file-server readme	298	F4CDD2EFC40FF49740EE7CAD118CB019E227BF119F24506194919E15380513CA
Incomplete HF mirror Parquet	301,458,815	05648F7B3761ED9B24A874FE08EE3BB62DAA6BAA0099D190AAC486E4737B360A
Mirror workbook	633,351	F8ABEABA6EEDF67869907B3A71EE132CD89B42BB3DCCE6B25DDB22E6269B51A7
DeepHS official repository	-	5f47812a3fb5bd8f0ed5367cedf9b464f83076c1
RipeTrack official repository	-	bf2a152eadfb322d55bb6345b34002e75fe97fb7

The machine-readable ledger is `reproducibility/source-checksums.csv`. The 77.3-GB DeepHS fruit payload and RipeTrack fruit packages were not downloaded because no explicit dataset reuse license was verified.

K.3 Determinism and split controls

- Fixed project seed: 20260730.
- Classification: five-fold shuffled StratifiedGroupKFold, group =

biological fruit.

- Regression: five-fold shuffled GroupKFold, group = biological fruit.
- Random-forest estimators and gradient-boosting estimators receive the fixed

seed.

- Fruit-cluster bootstrap: 1,000 iterations with fixed seeds.
- ml/results/split-audit.csv records zero fruit overlap in all ten folds.
- Predictor selection excludes fruit ID, filename, day, side, stage, and

storage condition.

- All out-of-fold predictions are retained, not reconstructed from summary

metrics.

Floating-point and parallel-tree results can vary at very small numerical precision across operating systems or library builds. Scientific conclusions must tolerate that variance; exact byte equality of all model artifacts is not promised.

K.4 Regeneration order

From a PowerShell prompt:

```
Set-Location -LiteralPath 'C:\Users\DELL\Desktop\Avocado'
& '.\reproducibility\run_all.ps1'
```

The runner:

1. checks that the immutable primary archive and extracted dataset exist;
2. sets the workspace-local analysis dependency path;
3. regenerates metadata and image features;
4. regenerates EDA tables/figures;
5. reruns fruit-grouped ML and uncertainty;
6. redraws the prototype overview;
7. validates tables, selected metrics, split overlap, required files, and source checksums.

-SkipLargeHashes skips only checksum calculation for files above 500 MB. It does not skip data/metric/split validation.

K.5 Automated validation

`reproducibility/validate\_outputs.py` checks:

- expected row counts and unique column names for 11 CSV artifacts;
- 14,722 source records, 14,710 images/matches, and 12 missing-image records;
- zero fruit overlap;
- the selected balanced-accuracy and remaining-time MAE values;
- required proposal/model files;
- recorded source SHA-256 values.

Its machine-readable output is reproducibility/validation-report.json. This is a structural and numerical check, not a substitute for scientific peer review or claim-boundary review.

K.6 PDF build and QA

The final PDF is built by:

```
$env:PYTHONPATH='C:\Users\DELL\Desktop\Avocado\tmp\python_deps'
& 'C:\Users\DELL\.cache\codex-runtimes\codex-primary-runtime\dependencies\python\python.exe' `
'..\tmp\pdfs\build_proposal_pdf.py'
```

The release process renders the PDF to page images with the bundled Poppler pdftoppm, inspects representative pages and montage/contact sheets, extracts text with pdfplumber, and checks page count, blank pages, and required phrases. All proposal dash characters are normalized to ASCII hyphens during PDF generation to avoid renderer-specific dash substitution.

K.7 Reproducibility boundaries

- The public RGB ground truth is an external composite ripening index, not instrumental chemistry.
- Dataset licenses remain source-specific.
- Current ML validation is one-study fruit-grouped validation.
- Storage holdout is a stress test, not an independent harvest.
- Candidate equations contain no fitted avocado rate constants.
- Hardware prices require re-quotation.
- A final prospective study requires preregistration, independent batches, device/chamber calibration, and locked external testing.

Appendix L - Risk, Evidence, Claims Boundaries, and Go/No-Go Gates

Decision status: pre-experimental planning **Evidence-audit date:** 30 July 2026 **Scope:** Hass avocado metrology, pilot experimentation, confirmatory study, modeling, and release decisions

L.1 Purpose

This appendix prevents a technically plausible prototype from becoming an unsupported scientific or deployment claim. It:

1. defines the evidence labels used throughout the project;
2. records what the present work does and does not establish;
3. links each material risk to the evidence that motivates it; and
4. defines staged go, conditional-go, and no-go decisions.

The gates are intentionally asymmetric. A failed optional sensor can be removed or relabeled without cancelling a well-designed experiment, but failed biological independence, endpoint validity, safety, data rights, or core metrology blocks the affected claim or phase.

L.2 Evidence and decision labels

Evidence labels

Label	Meaning	Permitted use
DATA-DEMONSTRATED	Directly reproduced from files inspected and analyzed in this workspace	May be reported with its dataset, validation unit, uncertainty, and scope
LITERATURE-SUPPORTED	Reported by a cited primary source but not independently reproduced here	May motivate a method or hypothesis; must retain the source's population and limitations
INFERRED	Reasoned interpretation of available evidence	May guide design; must not be presented as an observed result
PROPOSED	Future method, apparatus, equation, threshold, or analysis	Requires preregistration, calibration, and experimental validation
NOT IDENTIFIABLE	Current data cannot estimate the quantity, mechanism, or causal effect	The corresponding claim is prohibited until new evidence is collected

These definitions match the integrated proposal and the evidence corrections in Appendix A.

Decision labels

Label	Meaning
GO	All mandatory evidence for the phase exists and has passed its locked acceptance criteria
CONDITIONAL GO	Work may continue only with a narrower endpoint, downgraded sensor label, exploratory analysis, or documented limitation
NO-GO	Continuing would make the result unsafe, legally unsupported, scientifically uninterpretable, or non-reproducible
HOLD	A decision awaits external permission, a resource owner, a safety review, or a prespecified measurement

Numerical acceptance thresholds must be selected and signed before the data used to judge them are collected. They must not be chosen after viewing biological or locked-test outcomes.

L.3 Evidence anchors

Evidence anchor	What it establishes	Boundary
Dataset discovery and suitability audit	One located longitudinal avocado dataset is immediately reusable; RipeTrack and DeepHS are licence-blocked; image/patch count is not biological sample size	Does not establish performance outside the inspected datasets
Data-quality report	14,710 JPEGs match 478 fruits; 12 workbook rows have no corresponding image; labels are external visual/tactile stages	Does not provide firmness, moisture, oil, starch, pigment, or internal-quality truth
Fruit-grouped ML results	Best five-stage balanced accuracy is 0.731 internally; held-out T10 balanced accuracy falls to 0.497; days-to-stage-4 MAE is 1.853 days	Internal validation in one study; the time target is retrospective and not direct edible shelf life
Prototype architecture	Direct sensor observations, prohibited sensor claims, calibration controls, and eight readiness checks	Hardware selection does not itself validate biological meaning
Appendix A - supplied papers	RipeTrack feasibility and limitations; Bratu et al. gas method and mostly inconclusive imaging result	Literature findings are not replicated avocado results
Appendix C - dataset register	Access, licence, leakage, and recommended-use status for candidate datasets	Public download links without explicit rights remain blocked
Appendix D - hardware catalogue	Proposed parts, specifications, alternatives, and sourcing evidence	Prices, availability, and vendor claims require re-verification before purchase
Mendeley Hass dataset	CC BY 4.0 source of the reproduced RGB analysis	Controlled DSLR/lightbox domain, not uncontrolled smartphone deployment
RipeTrack	Published mobile RGB/NIR and remaining-life precedent	Only 13 independent avocado fruits, including three Hass; data licence unresolved
Bratu et al.	LPAS ethylene/ethanol workflow in apples	Apple gas trajectories are not avocado rate constants; imaging evidence was weak

L.4 Current claims and prohibited overclaims

Topic	Defensible current statement	Evidence label	Statement that must not be made
Public RGB corpus	The audited CC BY 4.0 corpus contains 14,710 available images from 478 Hass fruits under three storage groups	DATA-DEMONSTRATE D - quality report	"There are 14,710 independent avocado samples"
RGB stage prediction	Compact colour features carry information about the dataset's five-stage external ripening index under fruit-grouped validation	DATA-DEMONSTRATE D - ML results	"A camera can determine internal eating readiness"
Storage transfer	The selected colour model degraded materially when a storage group was held out; T10 balanced accuracy was 0.497	DATA-DEMONSTRATE D - ML results	"The RGB model generalizes across storage temperatures, seasons, or cameras"
Remaining life	A retrospective days-to-first-stage-4 target was modeled with 1.853-day MAE in internal fruit-grouped validation	DATA-DEMONSTRATE D - ML results	"The system predicts remaining edible shelf life to 1.853 days"
Peel colour biology	Chlorophyll loss and later anthocyanin accumulation contribute to Hass peel colour, and colour can desynchronize from softening	LITERATURE-SUPPORT - bibliography	"Darkness alone is a universal ripeness meter"
NIR feasibility	NIR can predict calibrated avocado properties such as dry matter under study-specific instruments and populations	LITERATURE-SUPPORT - bibliography	"A 970 nm LED directly measures internal moisture percentage"
970 nm channel	A repeatable 970 nm response may be retained as a water-sensitive feature after destructive calibration and ablation	PROPOSED - architecture	"The 970 nm signal is specific to water rather than scattering or other constituents"
Fruit mass	A calibrated load cell can estimate net mass change after tare, creep, and contact-force correction	PROPOSED - architecture	"All measured mass loss is water loss"
Ambient RH	A chamber sensor directly measures surrounding air temperature and RH	PROPOSED - architecture	"An RH sensor measures internal fruit water"
CO ₂	An in-range, leak-corrected NDIR accumulation slope can provide an apparent respiration-related feature	PROPOSED - architecture	"A single chamber CO ₂ value is total respiration without volume, time, leakage, and fruit-mass correction"
MOX gas sensor	A conditioned MOX response can be analyzed as a broad, humidity-sensitive VOC/odour trajectory	PROPOSED - architecture	"The BME688 measures ethanol or ethylene concentration"
Low-cost ethylene bridge	An electrochemical channel may be exploratory if compared with a reference method across relevant interferents	PROPOSED - architecture	"The low-cost sensor is selective quantitative ethylene in fruit headspace"
Governing equations	First-order, Arrhenius, logistic, Gompertz, state-space, and transport equations are candidate model families	PROPOSED - proposal	"The equations are established biological laws for this apparatus and cultivar"
Rate identification	Current public RGB data cannot identify water, pigment, respiration, oil, starch, or softening rate parameters	NOT IDENTIFIABLE - dataset audit	Any numerical mechanistic rate inferred from RGB labels alone
Dataset rights	The Mendeley Hass dataset has a verified CC BY 4.0 licence	DATA-DEMONSTRATE D - dataset record	"RipeTrack and DeepHS are reusable because their files are publicly linked"
Prototype status	The architecture and procurement catalogue define a buildable research prototype	PROPOSED	"The prototype is a validated consumer, clinical, food-safety, or commercial instrument"

L.5 Risk scoring

Likelihood and impact use **L** (low), **M** (medium), or **H** (high). A risk is **critical** when it can invalidate the main endpoint, create a safety or rights breach, or make the experimental unit unknowable. Ratings are planning judgments, not observed frequencies.

Scientific and biological risks

ID	Risk	L	I	Evidence or warning signal	Control and decision rule
SB-01	"Ready," "ripe," and "expired" remain ambiguously defined	H	H	Dataset labels are external stages, not universal eating quality: quality report	Stakeholder panel signs instrumental firmness, internal-condition, event, censoring, and error thresholds before confirmatory collection; otherwise NO-GO for remaining-useful-life claims
SB-02	Peel pigmentation and internal softening desynchronize	H	H	Storage holdout failure and pigment literature: ML results, bibliography	Retain colour only as a complementary channel; require firmness/internal reference; colour-only consumer readiness claim is NO-GO
SB-03	Harvest batch, season, supplier, temperature, or device shift causes failure	H	H	Held-out T10 balanced accuracy 0.497: ML results	Lock independent batches and report group-specific calibration; no independent cohort means CONDITIONAL GO for feasibility only
SB-04	Too few independent fruits, batches, or chambers produce unstable estimates	H	H	Public studies often have many images but few fruit: dataset audit	Prioritize independent fruit and batches over extra views; choose confirmatory n by blinded pilot simulation
SB-05	One chamber per condition confounds chamber with temperature/RH	M	H	Experimental hierarchy: proposal	Replicate chambers/cycles; otherwise label environmental effects exploratory and prohibit causal condition claims
SB-06	Matched sacrifice cohorts do not perfectly reveal a sentinel fruit's hidden state	M	H	Design limitation: proposal	Randomized matched cohorts, destructive windows, hierarchical uncertainty; explicitly retain approximation boundary
SB-07	Destructive assays are operator-, site-, or protocol-dependent	M	H	Required protocol controls: proposal	Freeze probe/site/speed/depth and chemical SOPs; duplicates and blinded QC; failed repeatability blocks that ground-truth claim
SB-08	Fruit attrition, rot, damage, or non-ripening biases time-to-event analysis	M	M	Censored source endpoints: quality report	Intention-to-observe population, reason codes, interval censoring, competing-defect reporting, missingness sensitivity

Metrology and hardware risks

ID	Risk	L	I	Evidence or warning signal	Control and decision rule
MH-01	NIR response confounds water, lipid/carbohydrate absorption, skin, geometry, and scattering	H	H	Sensor claims matrix and NIR literature: architecture, bibliography	Broad-spectrum reference subset, dark/white calibration, fixed geometry, destructive calibration and ablation; otherwise use only "optical feature" wording
MH-02	Low-cost gas sensors respond to interferents, temperature, RH, or drift	H	H	Cross-sensitivity boundary: architecture	Challenge gases and reference F-900/GC comparison; failure downgrades channel to nonspecific VOC trend or removes it
MH-03	CO ₂ saturation, leaks, or variable chamber volume biases the accumulation slope	M	H	Headspace controls: architecture	Empty/inert blanks, leak test, volume correction, range flags, shorter accumulation; out-of-range records are invalid
MH-04	Enclosure, tubing, adhesives, or printed parts outgas or adsorb volatiles	H	H	Mechanical limitations: architecture	Blank/recovery testing; inert wetted path for analytical work; persistent carry-over is NO-GO for compound claims
MH-05	LEDs, camera, fan, or condensation perturb fruit and sensor readings	M	M	Phase-separation requirement: architecture	Separate optical/headspace phases, fixed fan duty, thermal logging, recovery; excessive perturbation triggers redesign
MH-06	Scale creep, fan force, cradle contact, or repositioning creates false mass loss	M	M	Calibration requirements: architecture	Fan/pump off, isolated cradle, check masses, 24-hour creep and reposition tests
MH-07	Device-to-device or session calibration transfer fails	H	H	Controlled JPEGs lack absolute radiometric calibration: quality report	Device IDs, calibration records, transfer study, locked external device; no transfer evidence means device-specific claim only
MH-08	Sensor clocks or record keys are misaligned	M	H	File contract: architecture	One time standard, synchronization check, immutable session manifest; schema failure blocks biological collection
MH-09	Procurement substitution changes spectral, gas, or metrological behavior	M	M	Hardware catalogue	Re-quote and engineering review; no substitute is "equivalent" until range, accuracy, optical geometry, and interface are revalidated

Data, modeling, and reproducibility risks

ID	Risk	L	I	Evidence or warning signal	Control and decision rule
DM-01	Same fruit, source image, day, side, or patch appears across partitions	H	H	Published leakage examples: dataset audit	Immutable fruit IDs and automated zero-overlap audit; any overlap is NO-GO for reported test performance
DM-02	Retrospective endpoint construction leaks future information or changes the estimand	M	H	Days-to-stage-4 boundary: ML results	Define event and observation window prospectively; use survival/censoring methods; prohibit edible-life wording for retrospective external-stage targets
DM-03	Long-followed fruit dominate image-weighted metrics	H	M	T10 contributes about 60% of images but 40% of fruit: quality report	Fruit-cluster uncertainty, fruit-level summaries, storage-specific metrics, and weighting sensitivity
DM-04	Missing images, sensor failures, or assay failures are silently dropped	M	M	Twelve source records lack images: quality report	Immutable missingness flags, reason codes, population flow, sensitivity analyses; never synthesize replacements
DM-05	Hyperparameter tuning or feature selection sees the external cohort	M	H	Validation hierarchy: dataset audit	Nested/grouped tuning inside training fruits; external batch remains untouched until model freeze
DM-06	Flexible ODE/PINN/SINDy models return plausible but non-identifiable mechanisms	H	H	Identifiability boundary: proposal	Start with descriptive/mixed/state-space baselines; structural/practical identifiability, priors, perturbation validation; otherwise report prediction only
DM-07	Multiple sensor/biomarker screens create selective reporting	M	M	Statistical plan: proposal	Small primary family, hierarchical testing or FDR, effect sizes and intervals, publish null ablations
DM-08	Reproducibility fails because raw records or calibration lineage are overwritten	M	H	Data flow contract: architecture	Immutable raw layer, checksums, calibration and code versions, machine-readable manifests; provenance failure blocks release

Rights, safety, operational, and communication risks

ID	Risk	L	I	Evidence or warning signal	Control and decision rule
RS-01	Publicly linked data are treated as licensed	H	H	RipeTrack and DeepHS licence blocks: dataset audit, Appendix C	Acquire/use/redistribute only under explicit written terms; unresolved rights are NO-GO for that dataset
RS-02	Article, dataset, code, weights, and product images are assumed to share one licence	H	H	Rights distinctions: Appendix C	Maintain a separate source/licence ledger for every asset and derivative
RS-03	Solvents, compressed gases, heated ovens, or electrical hardware are used without institutional controls	M	H	Experimental methods and BOM: proposal, hardware catalogue	Institutional SOP, training, fume hood/ventilation, waste handling, electrical review; absent approval is NO-GO
RS-04	Budget or availability changes silently downgrade reference measurements	M	H	Staged procurement: proposal	Re-quote before purchase; protect ground truth and calibration first; narrow claims rather than substituting unvalidated parts
RS-05	Measurement burden changes the ripening process or causes unacceptable missingness	M	M	Repeated transfer warning: proposal	Metrology-stage timing study, shortened sessions, randomized order; excessive perturbation triggers redesign
RS-06	Results are communicated as deployment, food-safety, or commercial validation	M	H	Explicit limitations: proposal	Claims matrix, model card, uncertainty and abstention, review before external communication
RS-07	Contact details are guessed or an unverified address is used	M	M	No public email address verified for the intended recipient	Deliver through an established channel or verified LinkedIn profile; never infer an email address

L.6 Go/no-go gates

Gate 0 - collaborator fit and endpoint ownership

Evidence required

- Named scientific, statistical, engineering, data-governance, and laboratory responsibilities.
- Agreed primary use case and intended user.
- Draft operational definitions for ready-to-eat, end of useful life, internal defect, and actionable forecast error.
- Confirmation of access to at least two independent fruit batches for the pilot.
- Resource decision for firmness, moisture/dry matter, and any additional chemistry or reference gas claims.

GO: responsibilities, access, and endpoint-development process are accepted. **CONDITIONAL GO:** proceed with a narrower optical/environmental feasibility study if chemistry or reference gas facilities are unavailable; remove the affected internal-chemistry claims. **NO-GO:** no independent batches, no owner for ground-truth protocols, or no agreement on the primary endpoint.

Gate 1 - rights, safety, and procurement review

Evidence required

- Explicit licence or written permission for every external dataset, codebase, weight file, image, and redistributed asset.
- Current quotations and specification review for selected components.
- Institutional approval/SOPs for solvents, gas standards, ovens, electrical systems, waste, and any food handling.
- Approved materials for the headspace wetted path.

GO: rights and safety evidence are recorded in the source ledger and procurement review. **HOLD:** permission, quotation, or laboratory approval is pending. **NO-GO:** unresolved rights for a proposed dataset, unapproved chemical work, or an unsafe chamber/electrical design.

Gate 2 - metrology readiness before biological data collection

The prototype passes only when the numerical thresholds have been declared in advance and all of the following are documented:

1. 20 repeated scans of a stable optical reference satisfy the locked repeatability criterion;
2. fruit-removal and repositioning error is quantified for every scan zone;
3. mass calibration residuals, zero stability, and 24-hour creep pass;
4. CO₂ empty-chamber drift, leak rate, range compliance, recovery, and step response pass;
5. fixed camera settings and colour-target patches are reproducible across sessions;
6. clocks align within one logging interval;
7. chamber blank and VOC carry-over/recovery pass; and
8. every raw record validates against the data schema and retains calibration provenance.

Source: prototype acceptance criteria.

GO: all mandatory channels pass. **CONDITIONAL GO:** remove or relabel an optional failed modality and repeat its affected integration tests. Examples: a failed 970 nm calibration becomes a generic optical feature; a failed low-cost ethylene comparison becomes a nonspecific VOC feature or is removed. **NO-GO:** failure of fruit identity, clocks, raw-data provenance, camera/optical repeatability, mass integrity for a mass endpoint, chamber safety, or uncontrollable gas carry-over.

Gate 3 - pilot feasibility

Evidence required

- Metrology/feasibility work using stable controls and approximately 24-30 non-study fruits.
- A biological pilot using at least two independent receipt or harvest batches.
- Estimates of batch/chamber/fruit variance, missingness, endpoint frequency, censoring, assay repeatability, session burden, and candidate effect sizes.
- Evidence that the measurement process does not materially heat, dehydrate, bruise, or delay the fruit beyond the prespecified tolerance.
- Blinded simulation of confirmatory sample size and destructive windows.

GO: endpoints occur often enough, ground truth is repeatable, burden and missingness are manageable, and variance estimates support a feasible confirmatory design. **CONDITIONAL GO:** narrow conditions, remove a non-performing sensor, simplify assays, or change the primary target before preregistration. **NO-GO:** endpoint cannot be measured reproducibly, no independent batch effect can be estimated, or the apparatus materially changes the process it is meant to observe.

Gate 4 - confirmatory protocol lock

Evidence required

- Preregistered questions, hypotheses, primary endpoints, exclusions, censoring, multiplicity, model family, metrics, and actionable error thresholds.
- Sample size selected by pilot-based simulation; the 360-fruit layout in the proposal remains a planning design, not a fixed-power claim.
- Independent chamber replication or an explicit decision to treat condition effects as exploratory.
- Frozen fruit-grouped outer partitions and a locked independent batch/season/device cohort.
- Model and calibration freeze plan; no outcome-driven retuning on the external cohort.

GO: protocol, power simulation, external-cohort lock, and analysis code review are approved. **CONDITIONAL GO:** proceed as exploratory if chamber or batch replication is insufficient, with causal/generalization claims removed. **NO-GO:** endpoint or split can still change after outcomes are viewed, same fruit can cross partitions, or the external cohort is used for tuning.

Gate 5 - sensor and model adoption

Evidence required

- Comparison on identical fruit/batch folds against prior-only, RGB-only, and simple statistical baselines.
- Paired fruit/batch errors, calibration, prediction-interval coverage, subgroup results, and computational burden.
- Predeclared ablations: colour only, NIR only, environment/mass only, gas only, colour+NIR, and all available channels.
- Independent batch/device performance and explicit domain-shift failure analysis.
- For rate/mechanism claims: structural and practical identifiability plus intervention or perturbation validation.

GO: the added channel/model provides a meaningful, externally robust improvement with acceptable uncertainty and burden. **CONDITIONAL GO:** retain a channel for research/teaching or as a nonspecific predictive feature if it is repeatable but not chemically specific. **NO-GO:** improvement exists only under image-level leakage, internal folds, post hoc endpoint changes, or a non-identifiable mechanistic model.

Gate 6 - publication, data release, and deployment boundary

Evidence required

- Reproducibility audit from immutable raw data to figures and reported metrics.
- Rights review for every released file, image, model, and derivative.

- Model card/measurement statement defining intended use, excluded use, subgroup performance, domain limits, uncertainty, abstention, and failure modes.
 - Null results and failed modalities retained in the technical record.
 - Independent scientific review of claims against this appendix.
- GO:** release research results and permitted artifacts with evidence labels and limitations. **CONDITIONAL GO:** publish a feasibility or methods paper without consumer-readiness, chemistry, or deployment claims. **NO-GO:** advertise a food-safety, commercial sorting, universal avocado, phone-only, or internal-chemistry capability without independent prospective validation and the necessary regulatory/quality system.

L.7 Current decision snapshot

Area	Current state	Decision
Public RGB data	Licensed, acquired, checksummed, reconciled, and analyzed	GO for scoped reproducible RGB research
RGB deployment	Internal grouped results and storage-shift warning only	NO-GO for deployment claims
RipeTrack/DeepHS data	Public links; explicit dataset licences not verified	HOLD / NO-GO for acquisition or reuse until permission
Prototype design	Architecture and BOM prepared; metrology not yet executed	HOLD before biological collection
Low-cost ethylene	Proposed bridge with known cross-sensitivity	HOLD pending reference validation
Internal moisture/oil/starch/pigment claims	No matched experimental ground truth yet	NOT IDENTIFIABLE / NO-GO
Candidate differential models	Mathematically specified as hypotheses	GO for simulation and identifiability planning; NO-GO as discovered laws
Two-batch pilot	Proposed and awaiting collaborators/resources	HOLD pending Gate 0 and Gate 1

L.8 Decision record template

Every gate decision should be appended to the project ledger with:

```
decision_id:
date:
gate:
decision: GO | CONDITIONAL GO | NO-GO | HOLD
decision_owner:
reviewers:
evidence_files_or_links:
locked_thresholds:
failed_checks:
scope_or_claims_removed:
corrective_action:
next_review_date:
```

Silence or an undocumented hardware substitution is not a gate approval.